

Electrical Engineering Line-up

ASI Job #: 8673
Qty. (1)
Customer: Performance Seats, Spartanburg, SC
Package Release: ASAP
Hardware Due: ASAP
Panel Complete: ASAP
Runoff Date: 11/09/26
Fixture Description: Blackbird Automated Carbon Fiber Assembly Cell
PLC Enclosure: 42" tall x 37" wide x 12" deep ~ Saginaw Controls Pt. # SCE 42XEL3712P
 42" tall x 36" wide sub-plate ~ Saginaw Controls Pt. #SCE-42P36
HMI Enclosure: 20" tall x 20" wide x 8" deep ~ Rittal Pt. #1060250 (no CNC modifications)
Support Arm Kit: N/A (Tekno swivel as described)
PLC: Allen-Bradley Compact Logics – Part #1769-L33ER
HMI: (x2) Maple Systems 15" Smart Touchscreen – Part #CMT2158XV2
System IP List:

PLC	XXX.XXX.X.XX
HMI #1	XXX.XXX.X.XX
HMI #2	XXX.XXX.X.XX
North Valve Manifold	XXX.XXX.X.XX
North Vacuum Manifold 1	XXX.XXX.X.XX
North Vacuum Manifold 2	XXX.XXX.X.XX
South Valve Manifold	XXX.XXX.X.XX
South Vacuum Manifold 1	XXX.XXX.X.XX
South Vacuum Manifold 2	XXX.XXX.X.XX
Conveyor VFD	XXX.XXX.X.XX
M710-iC Robot	XXX.XXX.X.XX
Graco Glue System	XXX.XXX.X.XX

Cycle Start Style:



Cycle Time: Varies: 1500 parts per week @ 5 days per week @ 2 shifts per day
Reference Job: 8664 (base controls), 8688 (HMI), 8550 (robot, controller), 8807 (compression)

Fixture Outline:

Automated Solutions, Inc. will design and manufacture (1) turnkey Carbon Fiber Assembly Cell for the Ford Blackbird 1st Row back & cushion frames. This cell is an assembly process for securing a variety of carbon fiber parts, plastic brackets, and steel stampings into a completed seat back assembly using an adhesive dispensing system. Parts will be conveyed by operators throughout the cell while a robot is automatically dispensing the adhesive for each station.

Machine Home Condition:

1. Robot home & nozzle purged
2. Conveyor off
3. Clamp cylinders retracted
4. Press cylinder retracted
5. Part present sensors OFF
6. Vacuum off & pressure sensors off

Stage #1 Anticipated Sequence of Operations (in no particular order):

- Robot in home position; glue dispensing nozzle is full & ready for automated dispensing
- Left Backframe Sidemember Sequence of Operations – Step #1
 - o Operator loads Left Backframe Sidemember (#09) into lower nest
 - Part present sensors verify parts loaded (x1)
 - o Operator loads Left Backframe Reinforcement Bracket (#10) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
 - o Operator loads Left Backframe Bolster (#11) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
 - o Operator loads Sidemember Tab (#15) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
- Right Backframe Sidemember Sequence of Operations – Step #1
 - o Operator loads Right Backframe Sidemember (#12) into lower nest
 - Part present sensors verify parts loaded (x1)
 - o Operator loads Right Backframe Reinforcement Bracket (#13) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
 - o Operator loads Right Backframe Bolster (#14) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
 - o Operator loads Sidemember Tab (#15) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
- Upper Crossmember Sequence of Operations – Step #1
 - o Operator loads Upper Crossmember (#01) into lower nest
 - Part present sensors verify parts loaded (x1)
 - o Operator loads Reinforcement Bracket (#04) into upper nest
 - Vacuum turns on for part & pressure switch confirms presence (x1)
- Operator uses Opto-touch to initiate auto sequence for Stage #1 Glue Process
- Robot dispenses glue as required per each stations unique process
 - o Upper nest automatically lowers into position when its respective glue process is complete
 - o Upper nest holds position until desired cure time is achieved and automatically unclamps
- Process repeats until all stations for stage #1 are complete

Stage #2 Anticipated Sequence of Operations (in no particular order):

- Robot complete with Stage #1; glue dispensing nozzle is full & ready for automated dispensing
- Left Backframe Sidemember Sequence of Operations – Step #2
 - o **Driver Back Build:** Operator loads LH IB Recliner Bracket (#17) into lower nest
 - Part present sensors verify parts loaded (x1)
 - o **Passenger Back Build:** Operator loads RH OB Recliner Bracket (#18) into lower nest
 - Part present sensors verify parts loaded (x1)
 - o Operator loads completed Left Backframe Sidemember from Step #1 (#09, #10, #11, #15) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
- Right Backframe Sidemember Sequence of Operations – Step #2
 - o **Driver Seat Build:** Operator loads LH OB Recliner Bracket (#16) into lower nest
 - Part present sensors verify parts loaded (x1)
 - o **Passenger Seat Build:** Operator loads RH IB Recliner Bracket (#19) into lower nest
 - Part present sensors verify parts loaded (x1)
 - o Operator loads completed Right Backframe Sidemember from Step #1 (#12, #13, #14, #15) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)

- Lower Crossmember Sequence of Operations
 - o Operator loads Lower Crossmember (#07) into lower nest
 - Part present sensors verify parts loaded (x1)
 - o Operator loads Crossmember Bracket (#08) into upper nest
 - Vacuum turns on for part & pressure switch confirms presence (x1)
- Upper Crossmember Sequence of Operations – Step #2
 - o Operator loads completed Upper Crossmember from Step #1 (#01, #04) into lower nest
 - Part present sensors verify parts loaded (x1)
 - o Operator loads both Crossmember Tabs (#03) into their respective upper nests
 - Vacuum turns on for each part & pressure switch confirms presence (x2)
 - o Operator loads Plastic Strip (#02) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
 - o Operator loads Front Headrest (#05) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
- Operator uses Opto-touch to initiate auto sequence for Stage #2 Glue Process
- Robot dispenses glue as required per each stations unique process
 - o Upper nest automatically lowers into position when its respective glue process is complete
 - o Upper nest holds position until desired cure time is achieved and automatically unclamps
- Process repeats until all stations for stage #2 are complete
- Operator places all completed sub-assemblies from Stage #2 onto the belt conveyor to transfer the parts to the other side of the cell for Stage #3

Stage #3 Anticipated Sequence of Operations (in no particular order):

- Robot complete with Stage #1 & #2; glue dispensing nozzle full & ready for automated dispensing
- Operator retrieves all completed sub-assemblies from Stage #2 from the belt conveyor
- Upper Crossmember Sequence of Operations – Step #3
 - o Operator loads completed Upper Crossmember from Step #2 (#01, #02, #03, #04, #05) into lower nest
 - Part present sensors verify parts loaded (x1)
 - o Operator loads Rear Headrest (#06) into upper nest (headrest in pre-trim state)
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
 - o Automated glue dispensing occurs & press automatically advances into position until the desired cure time is achieved and automatically unclamps
- Cushion Pan Sequence of Operations – Step #1
 - o Operator loads Cushion Pan (#20) into lower nest
 - Part present sensors verify parts loaded (x1)
 - o Operator loads both Middle Suspension Tabs (#24) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x2)
 - o Operator load Left, Right & Center Plastic Strips (#21, #22, #23) into upper nests
 - Vacuum turns on for each part & pressure switch confirms presence (x3)
- Operator uses Opto-touch to initiate auto sequence for Stage #3 Glue Process
- Robot dispenses glue as required per each stations unique process
 - o Upper nest automatically lowers into position when its respective glue process is complete
 - o Upper nest holds position until desired cure time is achieved and automatically unclamps
- Process repeats until all stations for stage #3 are complete

Stage #4 Anticipated Sequence of Operations (in no particular order):

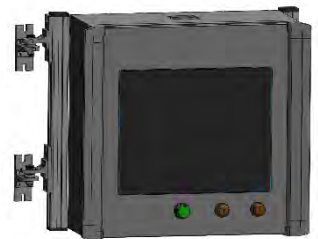
- Robot complete with Stage #1, #2 & #3; glue dispensing nozzle is full & ready for automated dispensing
- Cushion Pan Sequence of Operations – Step #2
 - o Operator loads completed Cushion Pan from Step #1 (#20, #21, #22, #23, #24) into lower nest
 - Part present sensors verify parts loaded (x1)
 - o Operator loads Left Cushion Pan Bolster (#26) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
 - o Operator loads Right Cushion Pan Bolster (#25) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)

- o **Driver Seat Build Only:**
 - Operator loads Left Suspension Tab (#27) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
 - Operator loads Right Suspension Tab (#28) into upper nest
 - Vacuum turns on for each part & pressure switch confirms presence (x1)
- o **Passenger Seat Build does not require installation of Left & Right Suspension Tabs**
- Backframe Final Assembly Sequence of Operations
 - o Operator loads completed Upper Crossmember from Step #4 (#01, #02, #03, #04, #5, #06) into nest (headrest in post-trim state)
 - Part present sensors verify parts loaded (x1)
 - o Operator loads completed Lower Crossmember (#07, #08) into nest
 - Part present sensors verify parts loaded (x1)
 - o Operator loads completed Left Sidemember from Step #2 (#09, #10, #11, #15, #17 or #18) into nest
 - Part present sensors verify parts loaded (x1)
 - o Operator loads completed Right Sidemember from Step #2 (#12, #13, #14, #15, #16 or #19) into nest
 - Part present sensors verify parts loaded (x1)
- Operator uses Opto-touch to initiate auto sequence for Stage #4 Glue Process
- Robot dispenses glue as required per each stations unique process
 - o Nest automatically lowers into position when its respective glue process is complete
 - o Nest holds position until desired cure time is achieved and automatically unclamps
- Process repeats until all stations for stage #4 are complete

Major components on machine:

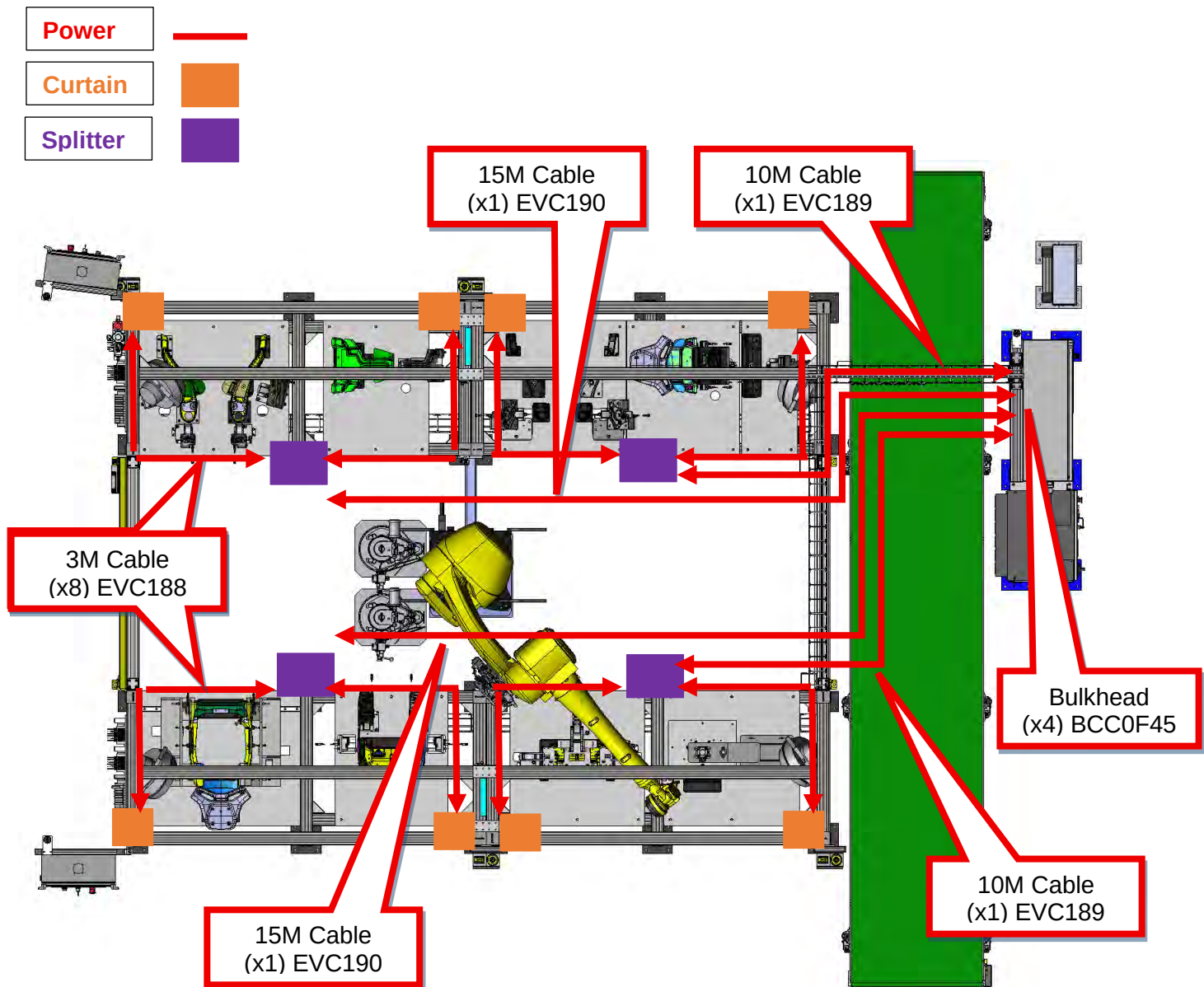
- 480VAC / 60A main power with 24VDC control voltage
 - o 42" tall x 37" wide x 12" deep ~ Saginaw Controls Pt. # SCE 42XEL3712P
 - o 42" tall x 36" wide sub-plate ~ Saginaw Controls Pt. #SCE-42P36
- (x1) 480VAC / 240VAC / 120VAC transformer ~ SolaHD Pt. # HS5F5AS
- (x1) 480VAC / 11A Robot ~ Fanuc Pt. #M-710iD_50M (PM71D50M-PAA1AU)
 - o (x1) Robot Controller #1 ~ Fanuc Pt. #R-30iB A Cabinet ~ includes pendant & cable
 - ID #: FXXXXXX ~ S/N: RXXXXXXXXX & EXXXXXXXXX ~ Date: 2026
 - o (x1) EtherNet/IP Adapter ~ Fanuc Pt. #RTL-R784
 - o (x1) Motion Package ~ Fanuc Pt. #RTL-R809
 - o (x1) Safety I/O Cable ~ Fanuc Pt. #CBLEO000000452O (#EWS-11613 DCS I/O CRMB2)
 - o (x1) Safety Package Advanced DCS ~ Fanuc Pt. #RTL-R859
 - o (x1) Software Package ~ Fanuc Pt. #RTL-R122
 - o Power, I/O & Communication to be hardwired with bulk cable as needed
 - o (x1) Dress Kit ~ TBD Pt. #TBD
- (x1) 220VAC / 10A Automated Glue Dispensing System ~ Graco Pt. #AP-260604-2
 - o **Itemize the assembly number above so that each component is listed for spare**
 - (x2) Check-Mate 20L Supply Pump ~ Graco Pt. #CM7C3F
 - (x1) Posi-Ratio Variable Dispensing System ~ Graco Pt. #PR-XV
 - (x1) Twin Control Mixing Valve ~ Graco Pt. #26D919
 - (x1) Static Mixing Valve ~ Graco Pt. #MCH 08-24T
 - (x1) Posi-Ratio Controller ~ Applied Adhesives Pt. #2002077
 - o Power, I/O & Communication to be hardwired with bulk cable as needed
 - o Adhesive Type: Plexus MA8105 with glass beads
 - o Adhesive Cure Time: 15min.
 - o Adhesive Working Time: 3-6 min. @ 75°, 5-7 min. @ 60°
 - o Adhesive Bead Size: 0.75mm to 2mm
 - o Purge Time: 3 min.
 - o Nozzle Diameter: 1/8"
 - o Nozzle Weight: 40# (includes meter, valve & shroud)
 - o Misc. EOAT Weight: 30# (includes hoses, material & mounting)

- (x1) 230VAC / 1.3A conveyor – Flex-Line Pt. #5AA.01.5050
 - o Motor driver required ~ Power & E-stop to come from main control panel
 - o Power to be hardwired with bulk cable as needed
- (x8) Overhead LED light required ~ Banner Pt. #WLB32ZC285MQMB
 - o Tie all to 5min timer relay but also include button on HMIs
 - o Maybe run power from HMI enclosures and tie into Harting connector?
- (x4) 18" Industrial fan ~ Airmaster Pt. #I-18YM (x1)
 - o Tie all to 5min timer relay but also include button on HMIs
 - o Maybe run power from HMI enclosures and tie into Harting connector?
- (x2) Metal Power Strip ~ Amazon Pt. #B0B54N4LVQX ~ (x1) per section
 - o Maybe run power from HMI enclosures and tie into Harting connector?
- (x4) E-stop Button ~ Banner Part #SSA-EB1PLYR-12ECQ8
 - o (x4) M12 8-Pin Female Bulkhead for enclosure ~ Balluff Pt. #BCC0E1L
 - o (x2) M12 8-Pin Female to M12 8-Pin Male 15M Cable ~ Balluff Pt. #BCC0PEC St. #3 & #4
 - o (x2) M12 8-Pin Female to M12 8-Pin Male 10M Cable ~ Balluff Pt. #BCC0FNW St. #1 & #2
 - o Mounted at each staging section
- (x4) Banner Opto Touch Button ~ Part #STBVR81Q6 (64192)
 - o (x4) IFM QD Cable for opto touch ~ Part #EVC114
 - o Mounted at each staging section
- (x1) Allen-Bradley Compact Logics PLC – Part #1769-L33ER
- (x2) Maple Systems 15" Smart Touchscreen ~ Pt. #CMT2158XV2
 - o (x2) Enclosure ~ Rittal Pt. #1060250 (includes sub-plate)
 - o (x2) Quick Cable Connector ~ Harting Pt. #TBD
 - o (x8) 1x1 Extrusion ~ Tekno Pt. #E0100201 x 19.50" Lg. (MBOM)
 - o (x8) 1x1 End Cap ~ Tekno Pt. #H0400221 (MBOM)
 - o (x4) Swivel Adapter ~ Tekno Pt. #B070051100 (MBOM)
 - o (x8) End Plate ~ Tekno Pt. #B0300121 (MBOM)
- (x3) Allen-Bradley Green System Start Button ~ Part #800T-QT24G
 - o Mounted on HMI panel door under HMI & main enclosure door
- (x2) Allen-Bradley Key Switch ~ Part #800T-H42A
 - o Mounted on HMI panel door under HMI
- (x1) Allen-Bradley E-Stop Button ~ Part #800T-TFXT6A4
 - o Mounted on main enclosure door
- (x1) Dual Ethernet programming port ~ Mencom Part # MEN-DP1-2RJ45-16LS
 - o Mount on side of enclosure
- (x2) Stack Light ~ Banner Pt. #TL50GRBAQ
 - o (x2) 5-Pin M12 Female 2M Cable ~ IFM Part #EVC070
 - o Mounted on top of HMI enclosure
 - o Maybe run power from HMI enclosures and tie into Harting connector?
- (x4) Honeywell Handheld Scanner ~ Pt. #PDF417 (Stryper)
 - o Be sure to include power supply, cables, adapters, etc. (cradle not required)
 - o (x1) Mounted at each staging section
- (x1) Safety Cage Operator Control and Safety Switch
 - o (x1) Gate Switch ~ Schmersal Pt. #AZM161ST-12/11RKA-024
 - (x1) M12 8-Pin Female Bulkhead for enclosure ~ Balluff Pt. #BCC0E1L
 - (x1) M12 8-Pin Female to M12 8-Pin Male 15M Cable ~ Balluff Pt. #BCC0PEC
 - (x1) 4-Pin M12 Female Bulkhead for enclosure ~ Balluff Pt. #BCC0CZ1
 - (x1) 4-Pin M12 Female to 4-Pin M12 Male 15M Cable ~ IFM Pt. #EVC117
 - o (x1) Switch Actuator ~ Schmersal Pt. #AZM161-B6
 - o (x1) Lockout Device ~ Schmersal Pt. #SZ-16/335
 - o (x1) Control Panel ~ Schmersal Pt. #BDF200-NH-11-LT-LT-LT-2875x2920
 - (x1) 12-Pin M12 Female Bulkhead for enclosure ~ Balluff Pt. #BCC0E5A
 - (x1) 12-Pin M12 Female to 12-Pin M12 Male 15M Cable ~ Balluff Pt. #BCC0HNR
- (x1) Safety Mat ~ Pinnacle Pt. #NSD-56-80-Y-W-20
- (x1) Safety Mat ~ Pinnacle Pt. #NSD-55-56-Y-W-20
- (x1) Mat Safety Relay ~ Pinnacle Pt. #NSD-DR-01
- (x1) Audible Alert ~ McMaster Carr Pt. # 56965T23



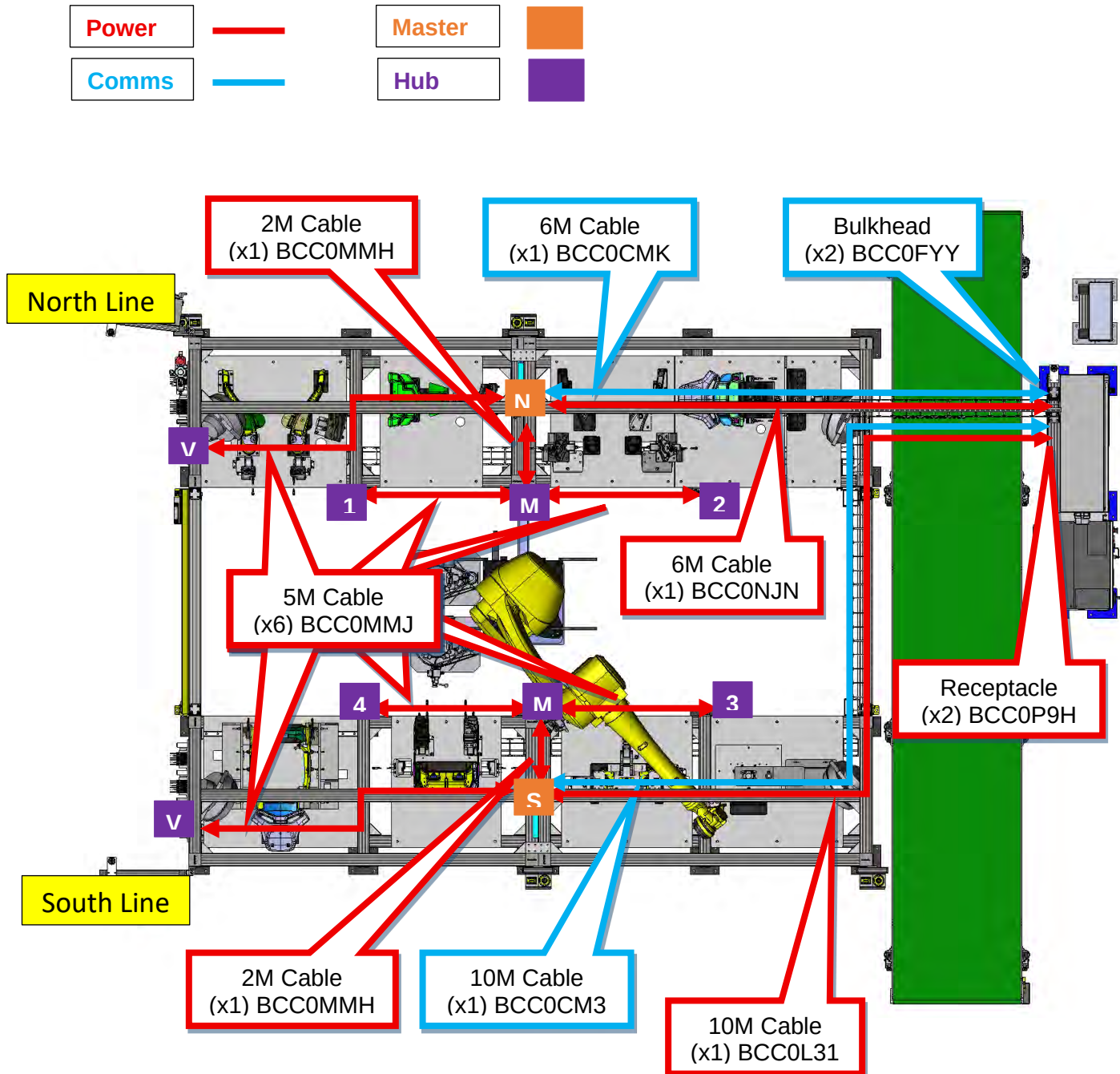
- (x1) Safety controller ~ Pt. #GC-1000
 - o (x1) Safety controller expansion unit ~ Pt. #GC-S84
 - o (x1) Safety controller expansion output relay ~ Pt. #GC-S1R
- (x4) Keyence Safety Light Curtain ~ Keyence Pt. #GL-VM48HP
 - o (x8) Mounting bracket ~ Pt. #GL-VB22
 - o (x4) M12 5-Pin Female to M12 5-Pin Male Bulkhead for enclosure ~ Balluff Pt. #BCC0F45
 - o (x4) Controller connection cable ~ Pt. #GL-VP2
 - o (x4) Y-Shaped Connector ~ Pt. #GL-VY01
 - o (x8) (x1) 5-Pin M12 Female to 5-Pin M12 Male 3M Cable ~ IFM Pt. #EVC188
 - o (x2) (x1) 5-Pin M12 Female to 5-Pin M12 Male 10M Cable ~ IFM Pt. #EVC189
 - o (x2) (x1) 5-Pin M12 Female to 5-Pin M12 Male 15M Cable ~ IFM Pt. #EVC190

Light Curtain Trunkline Schematic:



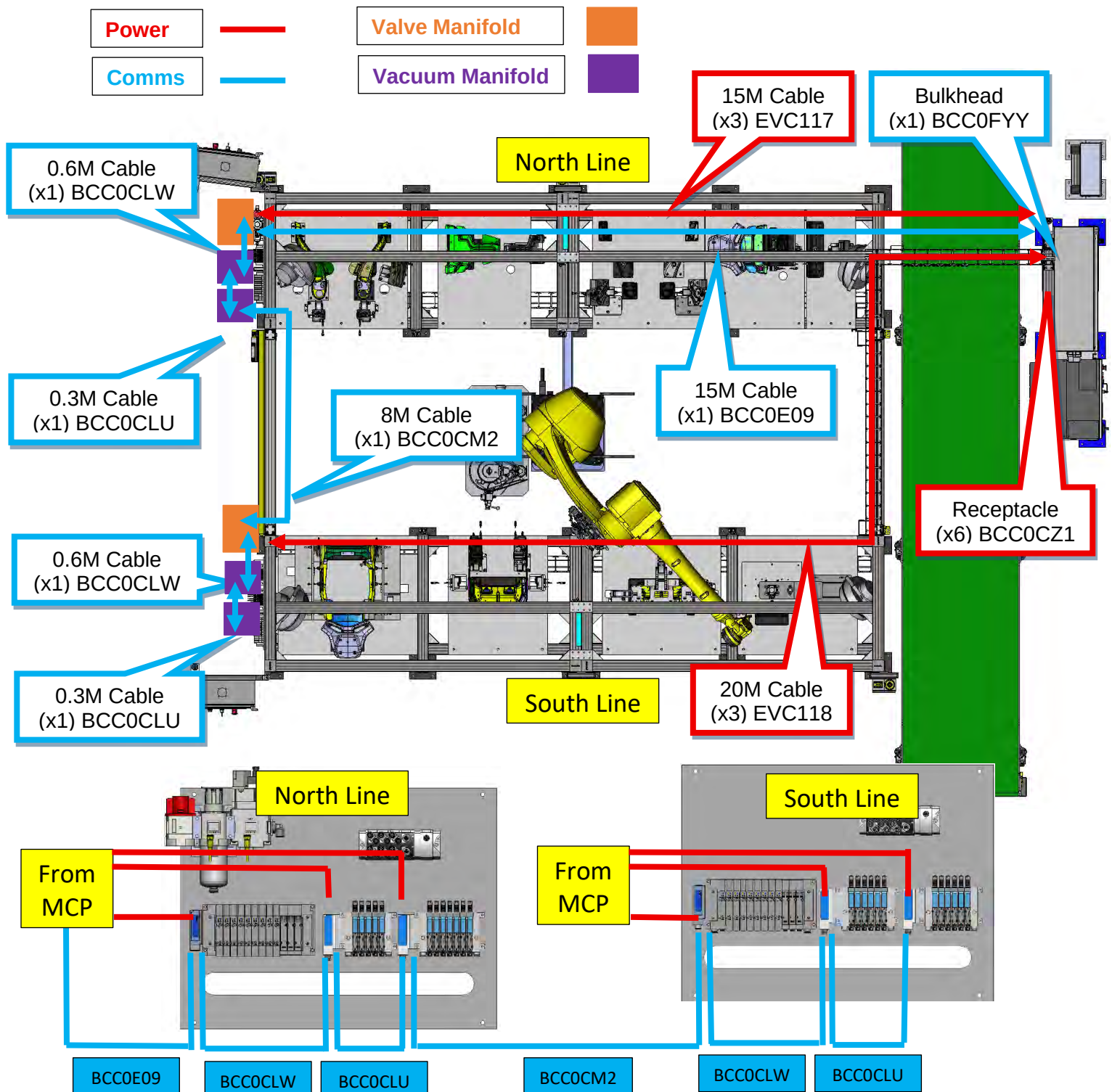
- I/O Block connections:
 - (x2) M12 4-Pin Female to RJ45 Male Bulkhead for enclosure ~ Balluff Pt. #BCC0FYF
 - (x1) M12 4-Pin Female to M12 4-Pin Male 10M Ethernet Cable ~ Balluff Pt. #BCC0CM3
 - (x1) M12 4-Pin Male to M12 4-Pin Male 6M Ethernet Cable ~ Balluff Pt. #BCC0CMK
 - (x2) M12 5-Pin Female Receptacle for enclosure ~ Balluff Pt. #BCC0P9H
 - (x1) M12 5-Pin Female to M12 5-Pin Male 6M Power Cable ~ Balluff Pt. #BCC0N3N
 - (x1) M12 5-Pin Female to M12 5-Pin Male 10M Power Cable ~ Balluff Pt. #BCC0L31
 - (x6) M12 4-Pin Female to M12 4-Pin Male 5M Power Cable ~ IFM Pt. #EVC014
 - (x2) M12 4-Pin Female to M12 4-Pin Male 2M Power Cable ~ IFM Pt. #EVC013

I/O Block Trunkline Schematic:



- Manifold connections:
 - o (x1) M12 4-Pin Female to RJ45 Male Bulkhead for enclosure ~ Balluff Pt. #BCC0FYY
 - o (x1) M12 4-Pin Female to M12 4-Pin Male 15M Ethernet Cable ~ Balluff Pt. #BCC0E09
 - o (x1) M12 4-Pin Female to M12 4-Pin Male 8M Ethernet Cable ~ Balluff Pt. #BCC0CM2
 - o (x2) M12 4-Pin Female to M12 4-Pin Male 0.6M Ethernet Cable ~ Balluff Pt. #BCC0CLW
 - o (x2) M12 4-Pin Female to M12 4-Pin Male 0.3M Ethernet Cable ~ Balluff Pt. #BCC0CLU
 - o (x6) M12 4-Pin Female Receptacle for enclosure ~ Balluff Pt. #BCC0CZ1
 - o (x3) M12 4-Pin Female to M12 4-Pin Male 20M Power Cable ~ IFM Pt. #EVC118
 - o (x3) M12 4-Pin Female to M12 4-Pin Male 15M Power Cable ~ IFM Pt. #EVC117

Manifold Trunkline Schematic:



Mechanical components for reference (on MBOM):

- (x1) Custom aluminum extrusion frame
- (x1) Robotic Standard Air Prep Assembly ~ SMC Pt. #FG25123L (MBOM)
 - o (x1) Filter Regulator with digital input gauge ~ SMC Pt. #AW40K-N04-Z-D
 - o (x1) Soft Start Valve with digital input gauge ~ SMC Pt. #AV4000-N04-5DZ-Z-D
 - o (x2) Digital Input Pressure Switch ~ SMC Pt. #ISE20B-Y-N01-W-X505
- (x1) North Line Manifold Sub-Assembly (cylinders only) (all on MBOM)
 - o (x1) 12-Station Valve Manifold ~ SMC Pt. #SS5Y5-10SEAN-12B-N7
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. #SY5400-5U1-NA **VALVE #1**
 - (x2) Left Sidemember Bolster & Bracket Clamp Cylinders
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. #SY5400-5U1-NA **VALVE #2**
 - (x2) Right Sidemember Bolster & Bracket Clamp Cylinders
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. #SY5400-5U1-NA **VALVE #3**
 - (x1) Reinforcement Bracket Clamp Cylinder
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. #SY5400-5U1-NA **VALVE #4**
 - (x1) Right Sidemember Recliner Clamp Cylinder
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. #SY5300-5U1-NA **VALVE #5**
 - (x1) Right Sidemember Up/Down Cylinder
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. #SY5400-5U1-NA **VALVE #6**
 - (x1) Left Sidemember Recliner Clamp Cylinder
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. #SY5400-5U1-NA **VALVE #7**
 - (x1) Left Sidemember Up/Down Cylinder
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. #SY5300-5U1-NA **VALVE #8**
 - (x1) Front Headrest Clamp Cylinder
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. #SY5400-5U1-NA **VALVE #9**
 - (x1) Crossmember Bracket Clamp Cylinder
 - (x3) Blank plate ~ Pt. #SY50M-26-1A-NA **VALVE #10, #11 & #12**
- (x1) North Line Stage #1 Vacuum Unit Sub-Assembly (generators only) (all on MBOM)
 - o (x1) 5-Station Vacuum Manifold ~ SMC Pt. #ZZK205A-AN1SEAN
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #1**
 - (x2) Left Sidemember Bolster Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #2**
 - (x1) Left Sidemember Bracket Cup
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #3**
 - (x2) Right Sidemember Bolster Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #4**
 - (x1) Right Sidemember Bracket Cup
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #5**
 - (x2) Reinforcement Bracket Cups
- (x1) North Line Stage #2 Vacuum Unit Sub-Assembly (generators only) (all on MBOM)
 - o (x1) 7-Station Vacuum Manifold ~ SMC Pt. #ZZK207A-AN1SEAN
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #1**
 - (x2) Left Sidemember Recliner Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #2**
 - (x2) Right Sidemember Recliner Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #3**
 - (x2) Front Headrest Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #4**
 - (x3) Upper Crossmember Strip Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #5**
 - (x1) Left Upper Tab Cup
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #6**
 - (x1) Right Upper Tab Cup
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #7**
 - (x1) Lower Crossmember Bracket Cup

- (x1) South Line Manifold Sub-Assembly (cylinders only) (all on MBOM)
 - o (x1) 12-Station Valve Manifold ~ SMC Pt. # SS5Y5-10SEAN-12B-N7
 - (x1) 2 pos. single solenoid valve ~ Pt. #SY5100-5U1-NA **VALVE #1**
 - (x1) Press Cylinder Brake Release
 - (x1) 3 pos. double sol. valve pressure center ~ Pt. #SY5500-5U1-NA **VALVE #2**
 - (x1) Press Cylinder
 - (x1) Sandwich regulator "A" side ~ Pt. #SY50M-N0-A
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. # SY5400-5U1-NA **VALVE #3**
 - (x1) Susp Brkts/Long Strip Clamp Cylinder & (x2) Side Strip Clamp Cyls.
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. # SY5400-5U1-NA **VALVE #4**
 - (x2) Bolster Clamp Cylinders
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. # SY5400-5U1-NA **VALVE #5**
 - (x2) Cushion Suspension Tab Clamp Cylinders (Driver Only)
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. # SY5400-5U1-NA **VALVE #6**
 - (x1) Upper Crossmember Clamp Cylinder
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. # SY5400-5U1-NA **VALVE #7**
 - (x1) Lower Crossmember Clamp Cylinder
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. # SY5400-5U1-NA **VALVE #8**
 - (x1) Left Sidemember Clamp Cylinder
 - (x1) 3 pos. double sol. valve exhaust center ~ Pt. # SY5400-5U1-NA **VALVE #9**
 - (x1) Right Sidemember Clamp Cylinder
 - (x3) Blank plate ~ Pt. #SY50M-26-1A-NA **VALVE #10, #11 & #12**
 - o (x1) 6-Station Vacuum Manifold ~ SMC Pt. #ZZK206A-AN1SEAN
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #1**
 - (x4) Rear Headrest Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #2**
 - (x4) Center Cushion Pan Strip Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #3**
 - (x3) Left Cushion Pan Strip Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #4**
 - (x3) Right Cushion Pan Strip Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #5**
 - (x2) Left Cushion Tab Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #6**
 - (x2) Right Cushion Tab Cups
- (x1) South Line Stage #4 Vacuum Unit Sub-Assembly (generators only) (all on MBOM)
 - o (x1) 6-Station Vacuum Manifold ~ SMC Pt. #ZZK206A-AN1SEAN
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #1**
 - (x2) Left Cushion Bolster Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #2**
 - (x2) Right Cushion Bolster Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #3**
 - (x2) Left Driver Cushion Tab Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #4**
 - (x2) Right Driver Cushion Tab Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #5**
 - (x4) Left Sidemember Cups
 - (x1) Vacuum generator #ZK2C15K5CCA-07 **VACUUM #6**
 - (x4) Right Sidemember Cups
- (x2) Pressure sensors required for air-prep
- (x31) Part present sensor
- (x24) Vacuum Ejector

- (x2) I/O-Link Master block ~ Pt. #BNI00K6
- (x8) I/O-Link Hub block ~ Pt. #BNI00CN
- (x33) Splitter ~ Pt. #BCC089P
- (x2) Electric Actuator ~ SMC Pt. #LEY100T9D-300BM
 - o (x1) Actuator cable ~ SMC Pt. #LE-CE-8-B
 - o (x1) Fieldbus for Actuator ~ SMC Pt. #JXC918-LEFS25REB-150 (put on EBOM)

Inputs:

- (1) Control Power On
- (2) Auto Mode Enable Key Switch
- (2) Air Pressure Switches
- (4) Cycle Start Buttons
- (24) Vacuum Pressure Switches
- (31) Part Present Sensors
- (43) Adv/Ret Cylinder Switches
- (4) Light Curtain
- (2) Safety Mat
- (1) E-stop Relay

114 Inputs

Outputs:

- (1) Control Power On Button Light
- (36) Adv/Ret Valves
- (12) Adv/Ret Spare Valves
- (24) Vacuum Ejectors
- (10) Stack Lights

83 outputs

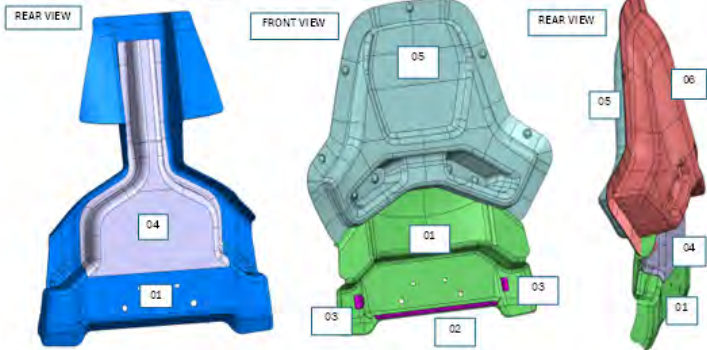
Station Shown with Notes/Comments:

- ASI standard hardware and controls practices
- LED light board not required
- Vision not required
- Torque equipment not required
- Alarm if glue dispenser is jammed or empty
- Use bulkhead & Harting connectors for all connections going back to main panel
- Please ensure 30% spare I/O
- Leave valves energized in their work and home states
- Lockout stickers will be required P1, E1
- Ensure all PP are labeled with I/O address

FSB Seat Components for Cell (see excel file for detailed information):

Part	Description	Material	Glue Area (in² 2)
1	Upper Crossmember	Carbon Fiber	27.86
2	Upper Crossmember Strip	Plastic	3.90
3	Upper Crossmember Tab (x2)	Steel	0.62
4	Upper Crossmember Reinforcement	Carbon Fiber	16.86
5	Headrest - Front (shown pre-trim)	Carbon Fiber	14.78
6	Headrest - Rear (shown pre-trim)	Carbon Fiber	69.60
7	Lower Crossmember	Carbon Fiber	10.14
8	Lower Crossmember Bracket	Steel	3.24
9	Sidemember - Left	Carbon Fiber	N/A
10	Sidemember Reinforcement - Left	Steel	11.29
11	Sidemember Bolster - Left	Carbon Fiber	13.60
12	Sidemember - Right	Carbon Fiber	N/A
13	Sidemember Reinforcement - Right	Steel	11.29
14	Sidemember Bolster - Right	Carbon Fiber	13.60
15	Sidemember Tab (x2)	Steel	0.62
16	LH Recliner Bracket - OB	Steel	15.63
17	LH Recliner Bracket - IB	Steel	N/A
18	RH Recliner Bracket - OB	Steel	15.63
19	RH Recliner Bracket - IB	Steel	N/A
20	Sidemember Bolster Flange - Left	Carbon Fiber	5.11
21	Sidemember Bolster Flange - Right	Carbon Fiber	5.11

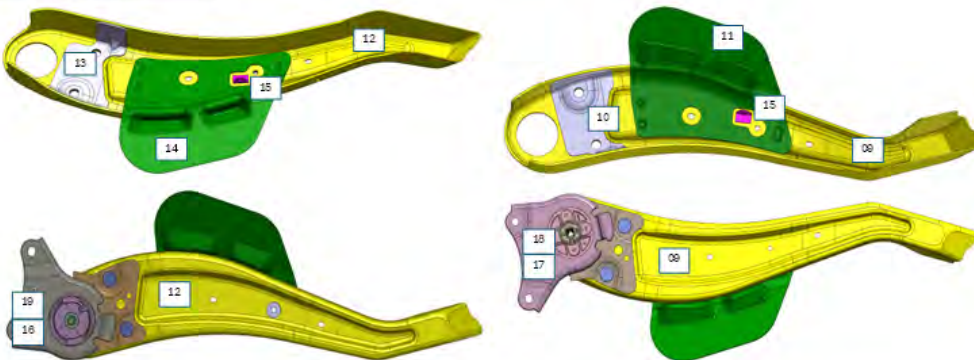
Upper Crossmember Glue Process



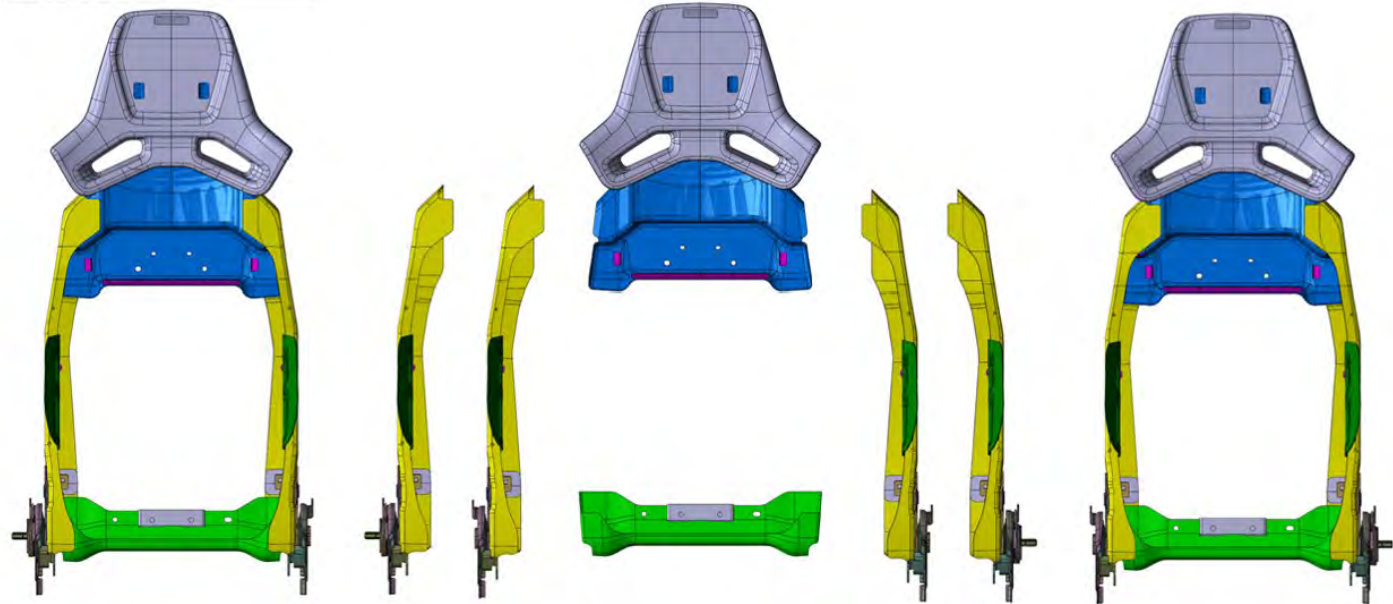
Lower Crossmember Glue Process



Sidemember Glue Process



Backframe Glue Process



Cushion Pan Seat Components for Cell (see excel file for detailed information):

Part	Description	Material	Glue Area (in^2)
20	Cushion Pan	Carbon Fiber	N/A
21	Cushion Pan Strip - Left	Plastic	1.77
22	Cushion Pan Strip - Center	Plastic	3.91
23	Cushion Pan Strip - Right	Plastic	1.77
24	Cushion Suspension Tab (x2)	Steel	3.04
25	Cushion Bolster - Left	Carbon Fiber	9.59
26	Cushion Bolster - Right	Carbon Fiber	9.59
27	Driver Cushion Suspension Tab - Left	Steel	1.24
28	Driver Cushion Suspension Tab - Right	Steel	1.24

Carbon Fiber Parts: 3

Steel Parts: 4

Plastic Parts: 3

Total # Parts: 10

Note:

Items #27 & #28 are only required for driver seats (not passenger)

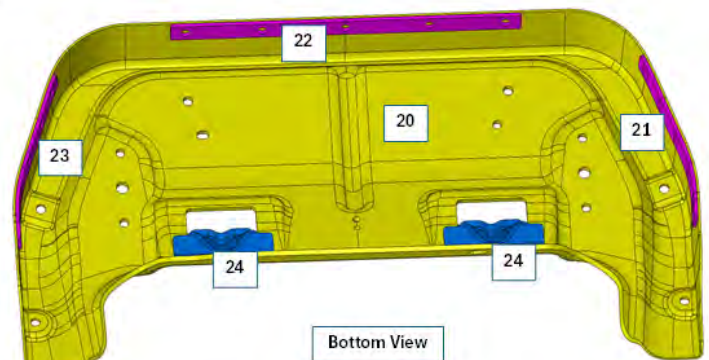
Total Part Qty. 10 for Driver & Qty. 8 for Passenger

Driver Pan Total Glue Area (in^2)	32.15
Passenger Pan Total Glue Area (in^2)	29.67
Driver Pan Squeeze Out Area (in^2)	3.22
Passenger Pan Squeeze Out Area (in^2)	2.97
Driver Total Glue Area (in^2)	35.37
Passenger Total Glue Area (in^2)	32.64

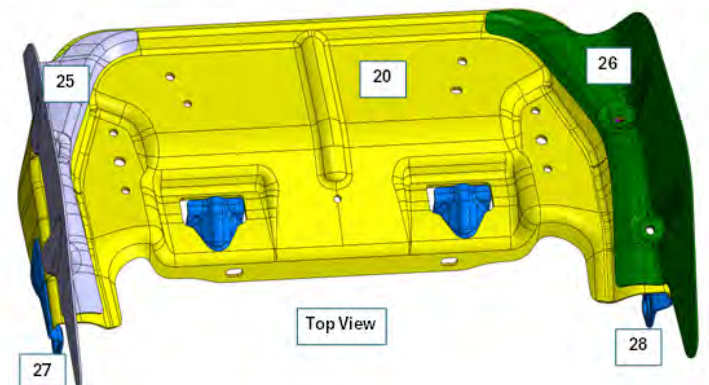
Nozzle Size (in.) 0.13

Bead Thickness (in) 0.08

Bottom Side Glue Process

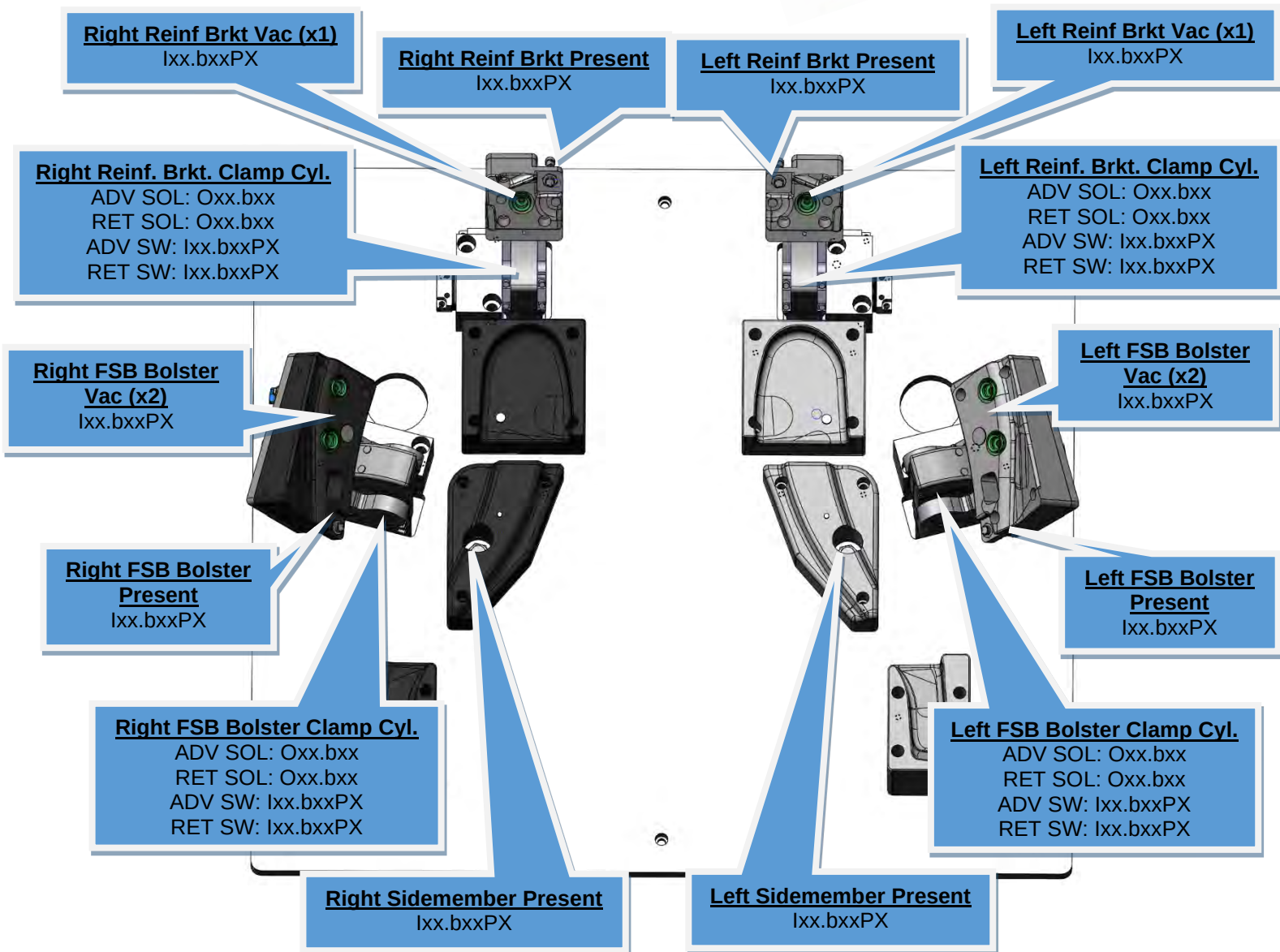
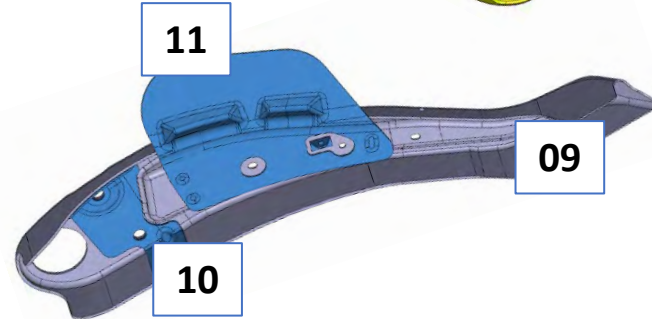
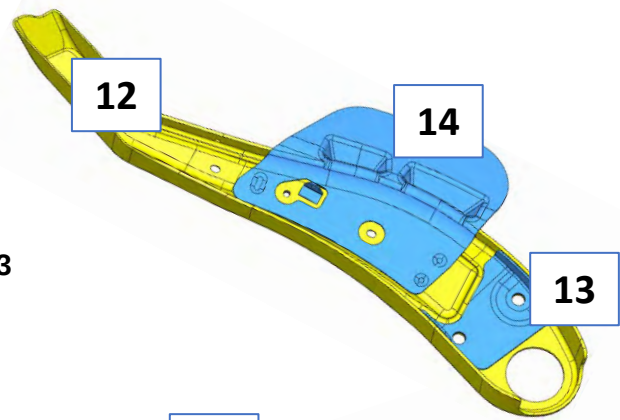


Top Side Glue Process

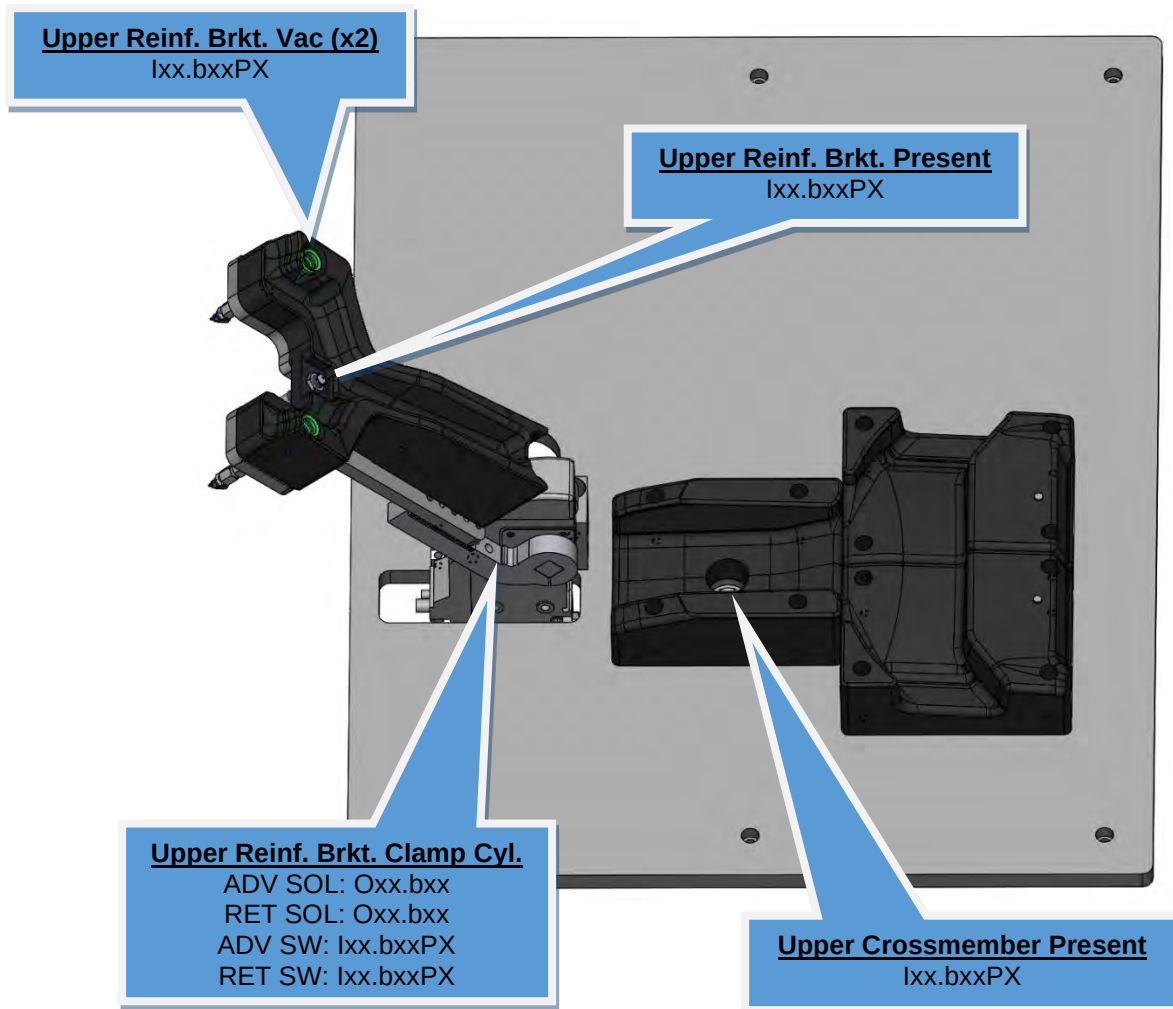
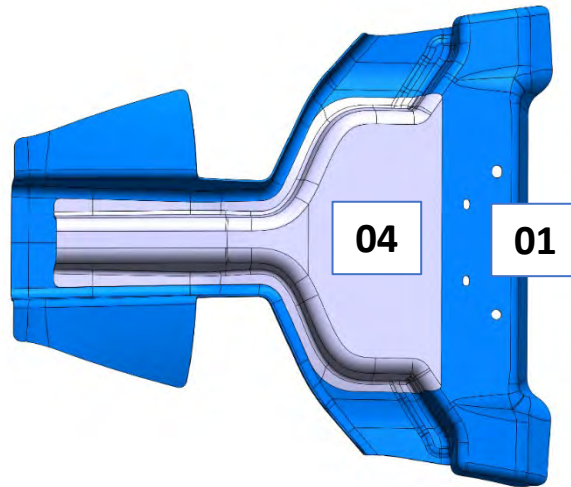


Stage #1 Nest Tooling:

- 8673-100 Right Sidemember:
 - o (x1) Right sidemember part present #12
 - o (x1) Right RSB bolster part present #14
 - o (x2) Right FSB bolster vacuum cups
 - o (x1) Right FSB bolster clamp cylinder
 - o (x1) Right reinforcement bracket part present #13
 - o (x1) Right reinforcement bracket vacuum cup
 - o (x1) Right reinforcement bracket clamp cylinder
- 8673-100 Left Sidemember:
 - o (x1) Left sidemember part present #09
 - o (x1) Left FSB bolster part present #11
 - o (x2) Left FSB bolster vacuum cups
 - o (x1) Left FSB bolster clamp cylinder
 - o (x1) Left reinforcement bracket part present #10
 - o (x1) Left reinforcement bracket vacuum cup
 - o (x1) Left reinforcement bracket clamp cylinder



- 8673-200 Upper Crossmember:
 - o (x1) upper crossmember part present #01
 - o (x1) upper crossmember reinforcement bracket part present #04
 - o (x2) upper crossmember reinforcement bracket vacuum cups
 - o (x1) upper crossmember reinforcement bracket clamp cylinder



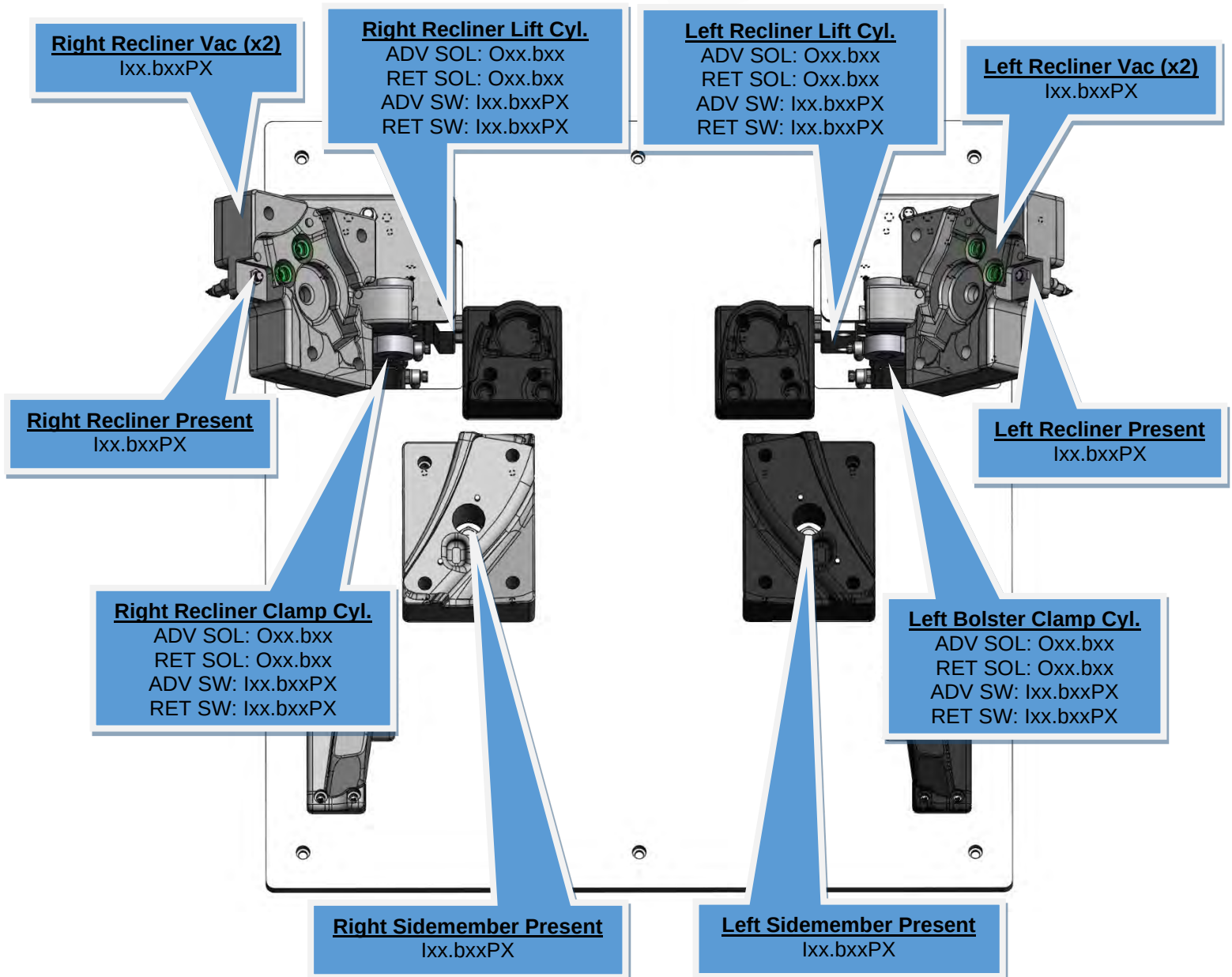
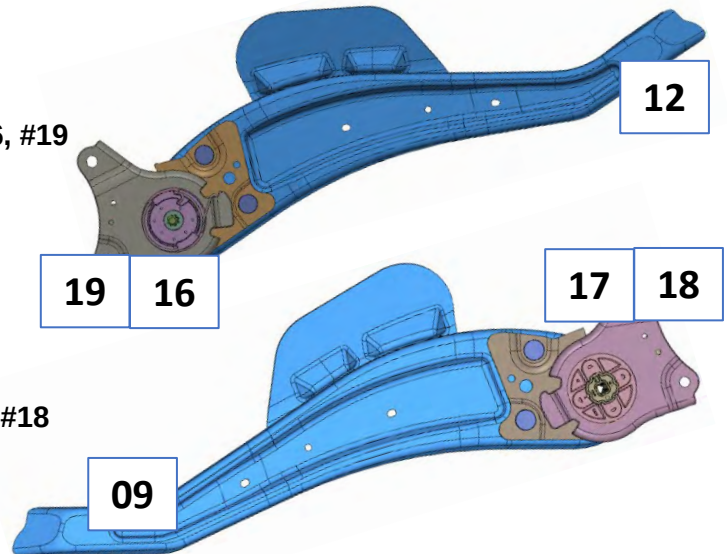
Stage #2 Nest Tooling:

- 8673-300 Right Sidemember:

- o (x1) Right sidemember part present #12
- o (x1) Right recliner bracket part present #16, #19
- o (x2) Right recliner bracket vacuum cups
- o (x1) Right recliner bracket clamp cylinder
- o (x1) Right clamp vertical lift cylinder

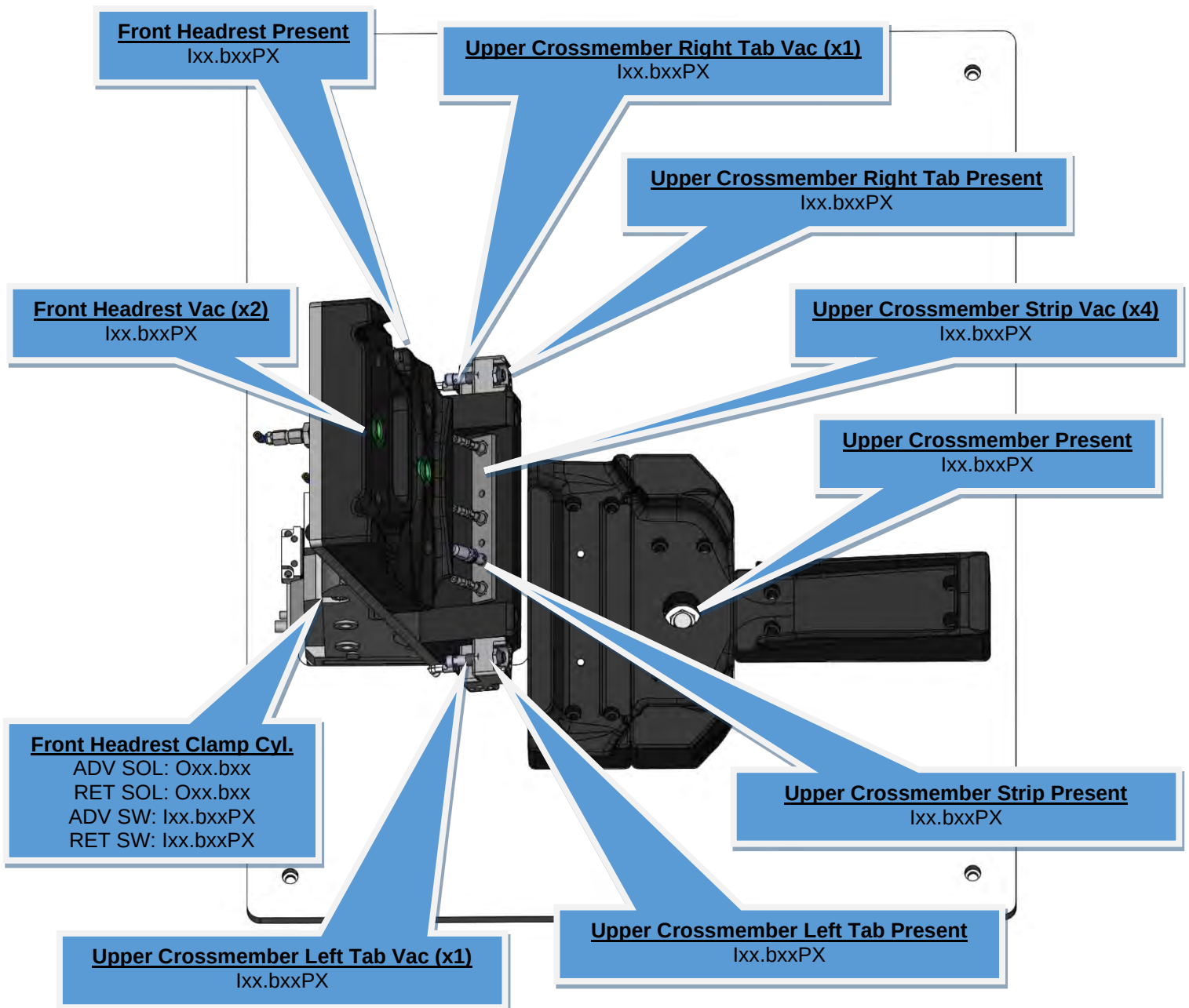
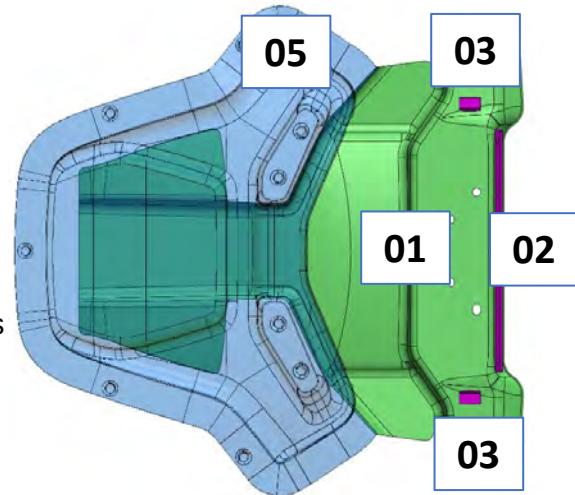
- 8673-300 Left Sidemember:

- o (x1) Left sidemember part present #09
- o (x1) Left recliner bracket part present #17, #18
- o (x2) Left recliner bracket vacuum cups
- o (x1) Left recliner bracket clamp cylinder
- o (x1) Left clamp vertical lift cylinder

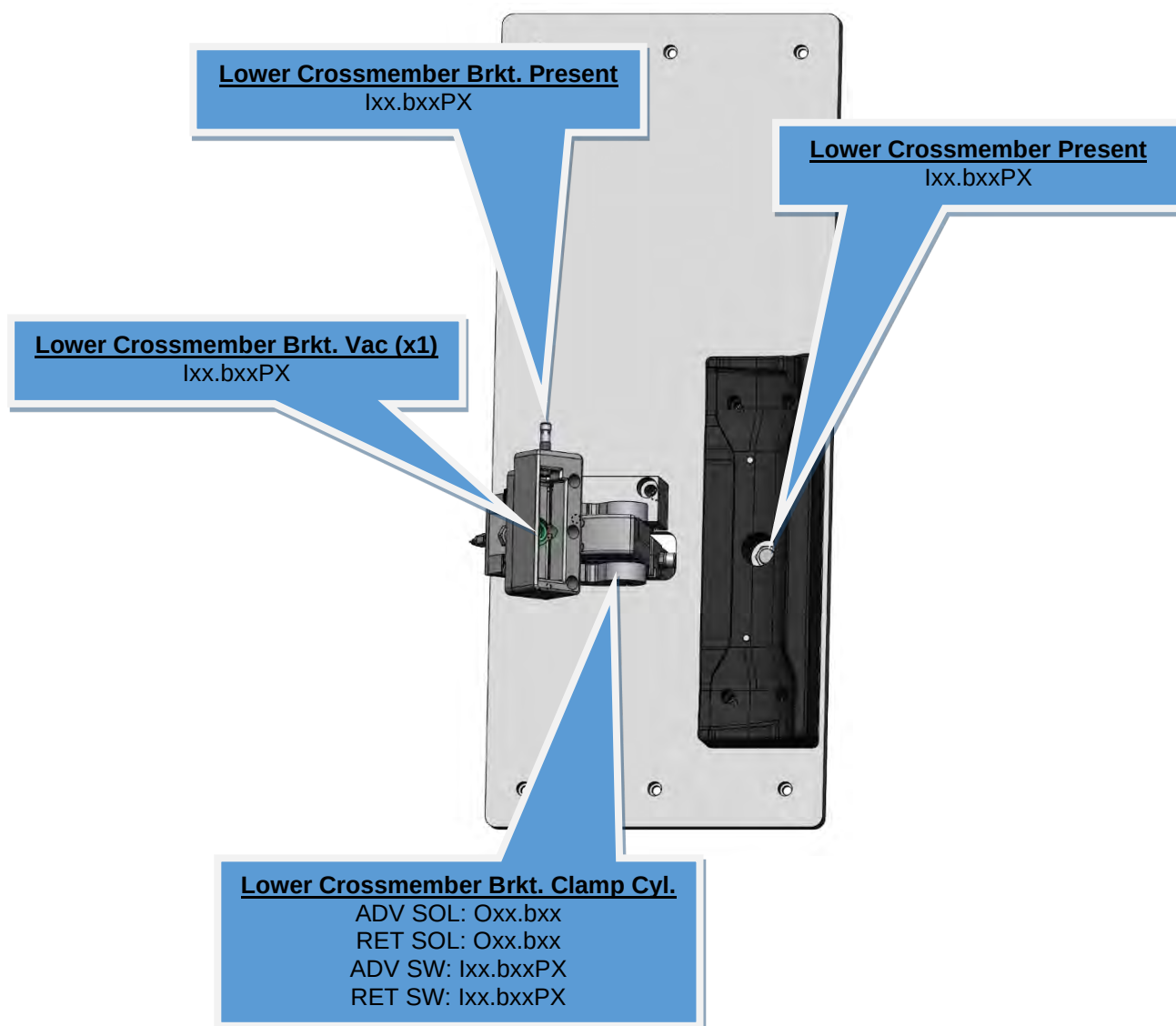
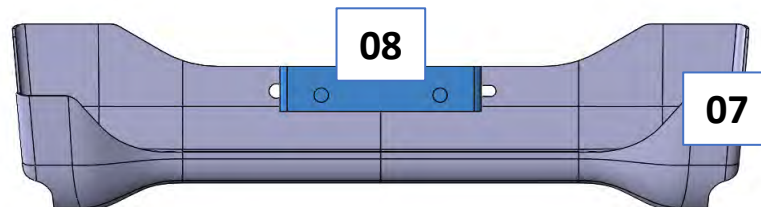


- 8673-400 Upper Crossmember:

- o (x1) upper crossmember part present #01
- o (x1) front headrest part present #05
- o (x2) front headrest vacuum cups
- o (x1) upper crossmember Left tab part present #03
- o (x1) upper crossmember Left tab bracket vacuum cups
- o (x1) upper crossmember Right tab part present #03
- o (x1) upper crossmember Right tab bracket vacuum cups
- o (x1) upper crossmember strip part present #02
- o (x3) upper crossmember strip vacuum cups
- o (x1) front headrest clamp cylinder

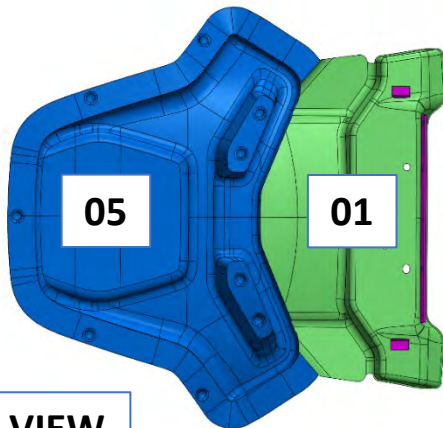


- 8673-500 Lower Crossmember:
 - o (x1) lower crossmember part present #07
 - o (x1) lower crossmember bracket part present cup #08
 - o (x1) lower crossmember bracket vacuum cup
 - o (x1) lower crossmember bracket clamp cylinder

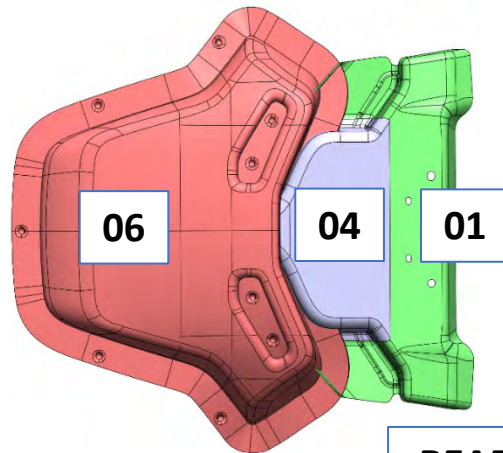


Stage #3 Nest Tooling:

- 8673-600 Upper Crossmember (Press Station):
 - o (x1) headrest part present (front) #05
 - o (x1) headrest part present (rear) #06
 - o (x4) headrest vacuum cups (rear)
 - o (x1) headrest press cylinder with brake



FRONT VIEW



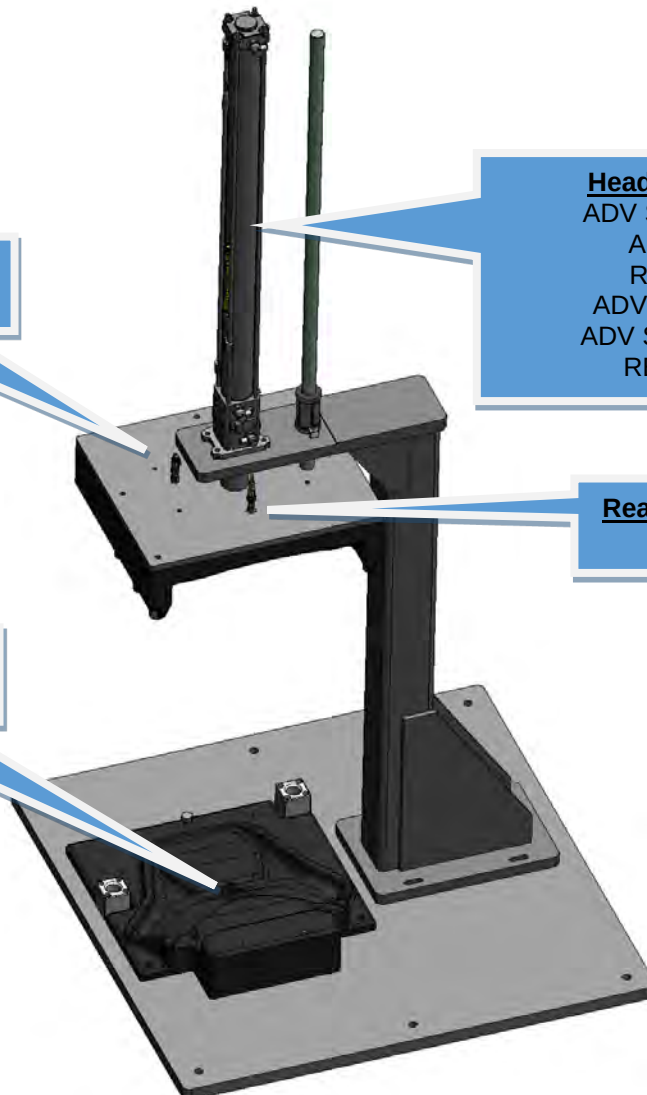
REAR VIEW

Rear Headrest. Vac (x4)
lxx.bxxPX

Headrest Press Cylinder
ADV SOL: Oxx.bxx (brake)
ADV SOL: Oxx.bxx
RET SOL: Oxx.bxx
ADV SW: lxx.bxxPX (full)
ADV SW: lxx.bxxPX (short)
RET SW: lxx.bxxPX

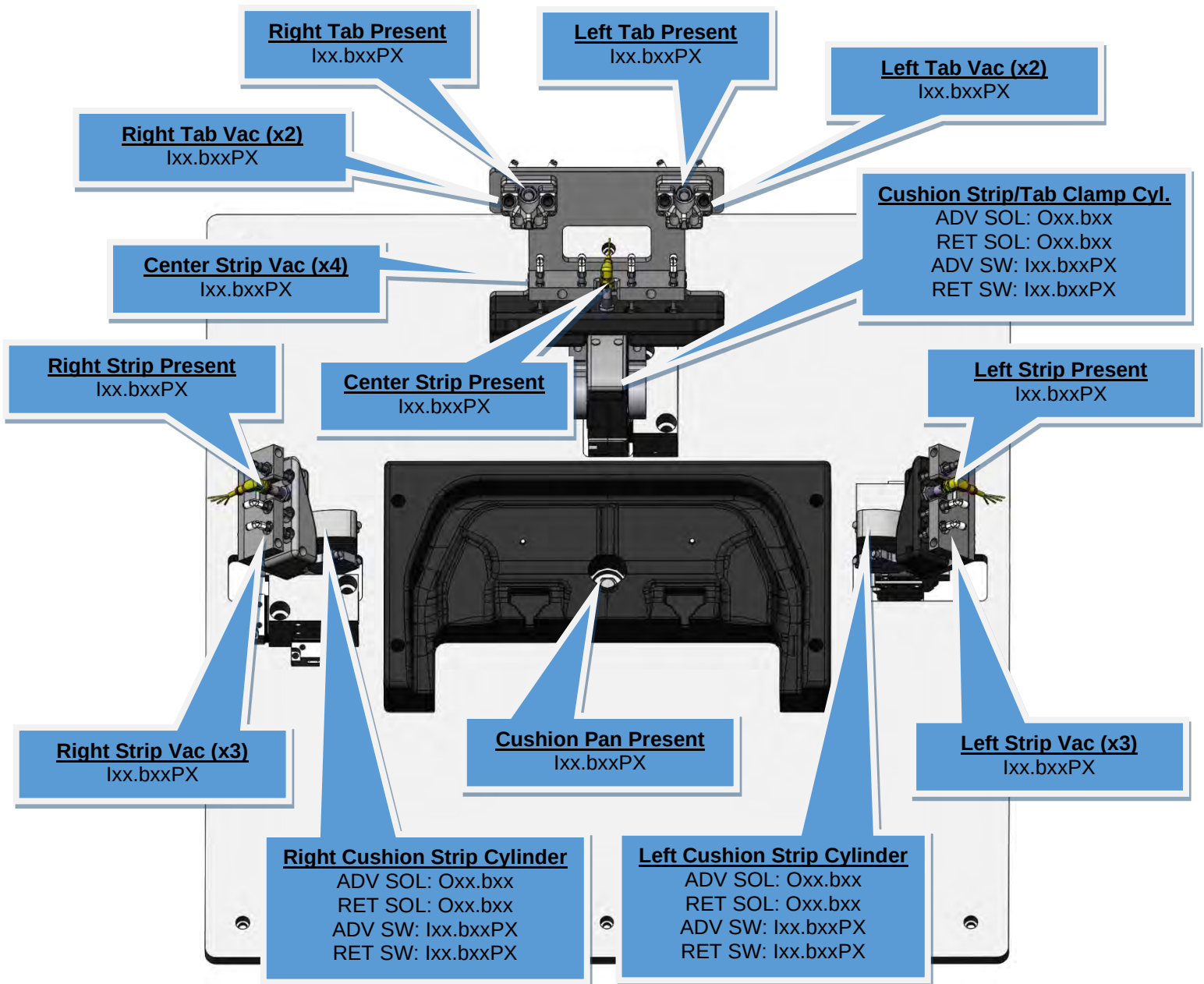
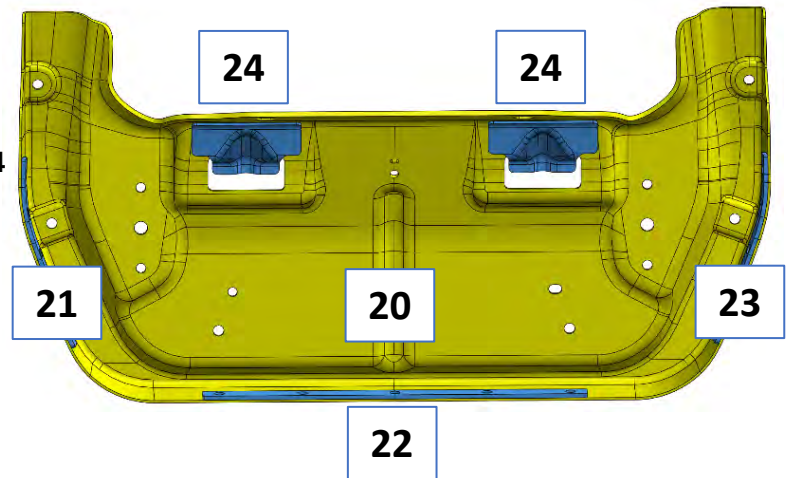
Rear Headrest. Present
lxx.bxxPX

Front Headrest. Present
lxx.bxxPX



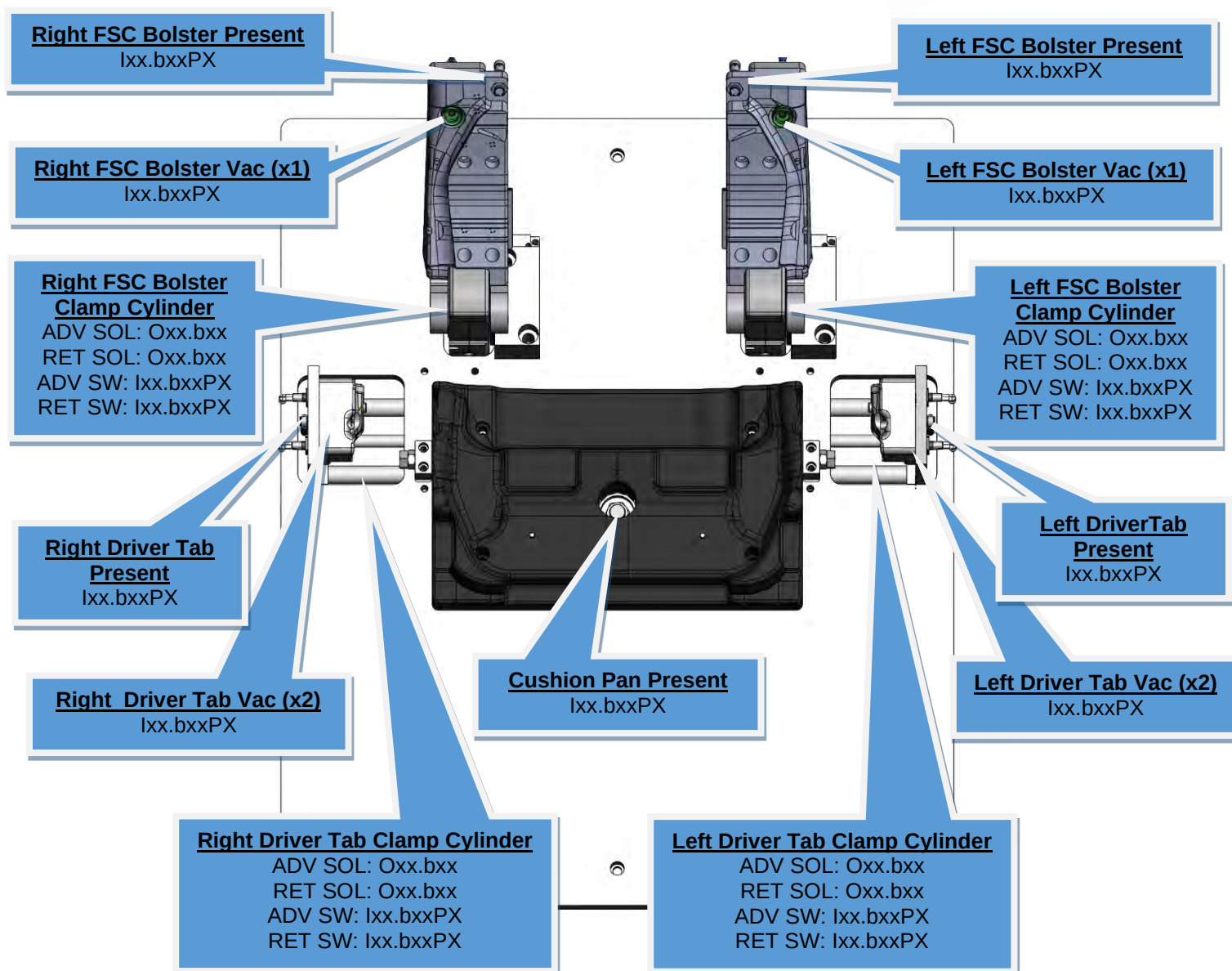
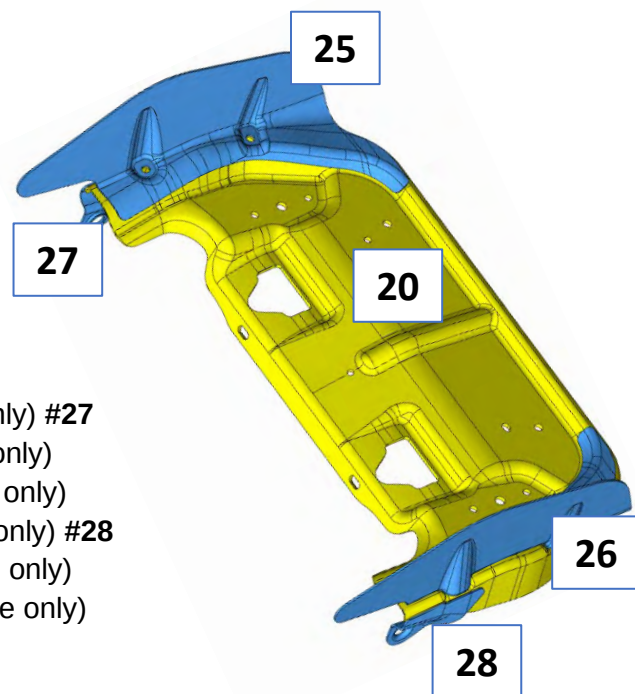
- 8673-700 Cushion Pan:

- o (x1) cushion pan part present #20
- o (x1) center strip part present #22
- o (x4) center strip vacuum cups
- o (x1) cushion Left tab part present #24
- o (x2) cushion Left tab vacuum cups
- o (x1) cushion Right tab part present #24
- o (x2) cushion Right tab vacuum cups
- o (x1) center strip & tabs clamp cylinder
- o (x1) side Left strip part present #21
- o (x3) side Left strip vacuum cups
- o (x1) side Left strip clamp cylinder
- o (x1) side Right strip part present #23
- o (x3) side Right strip vacuum cups
- o (x1) side Right strip clamp cylinder

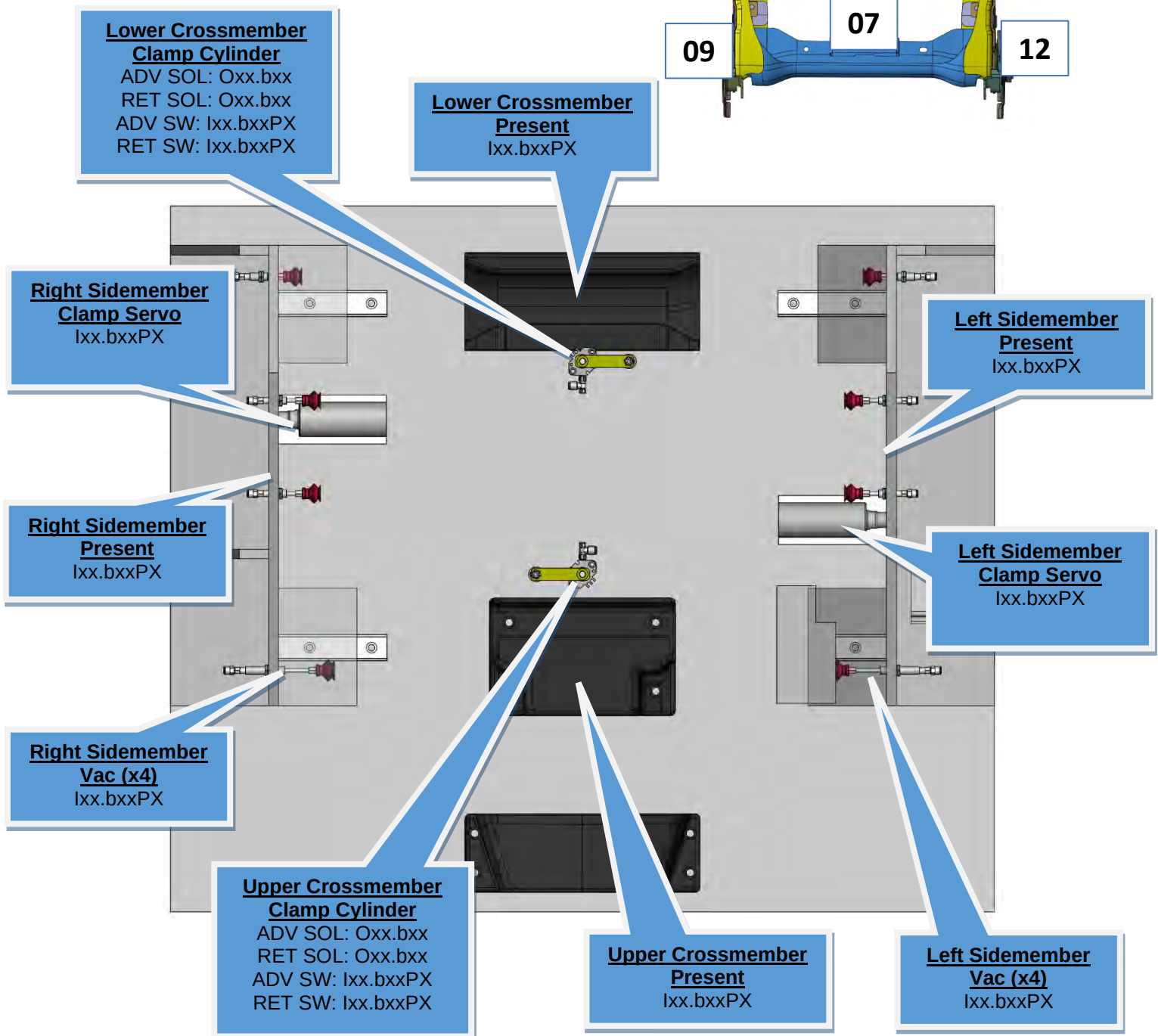
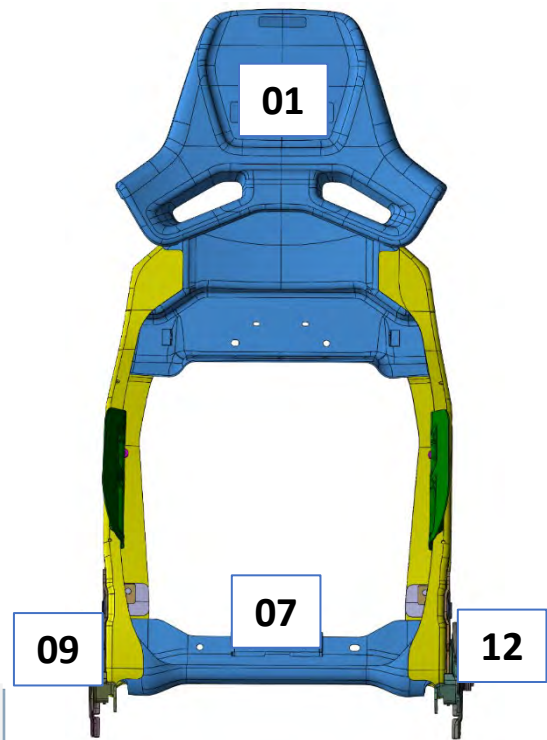


Stage #4 Nest Tooling:

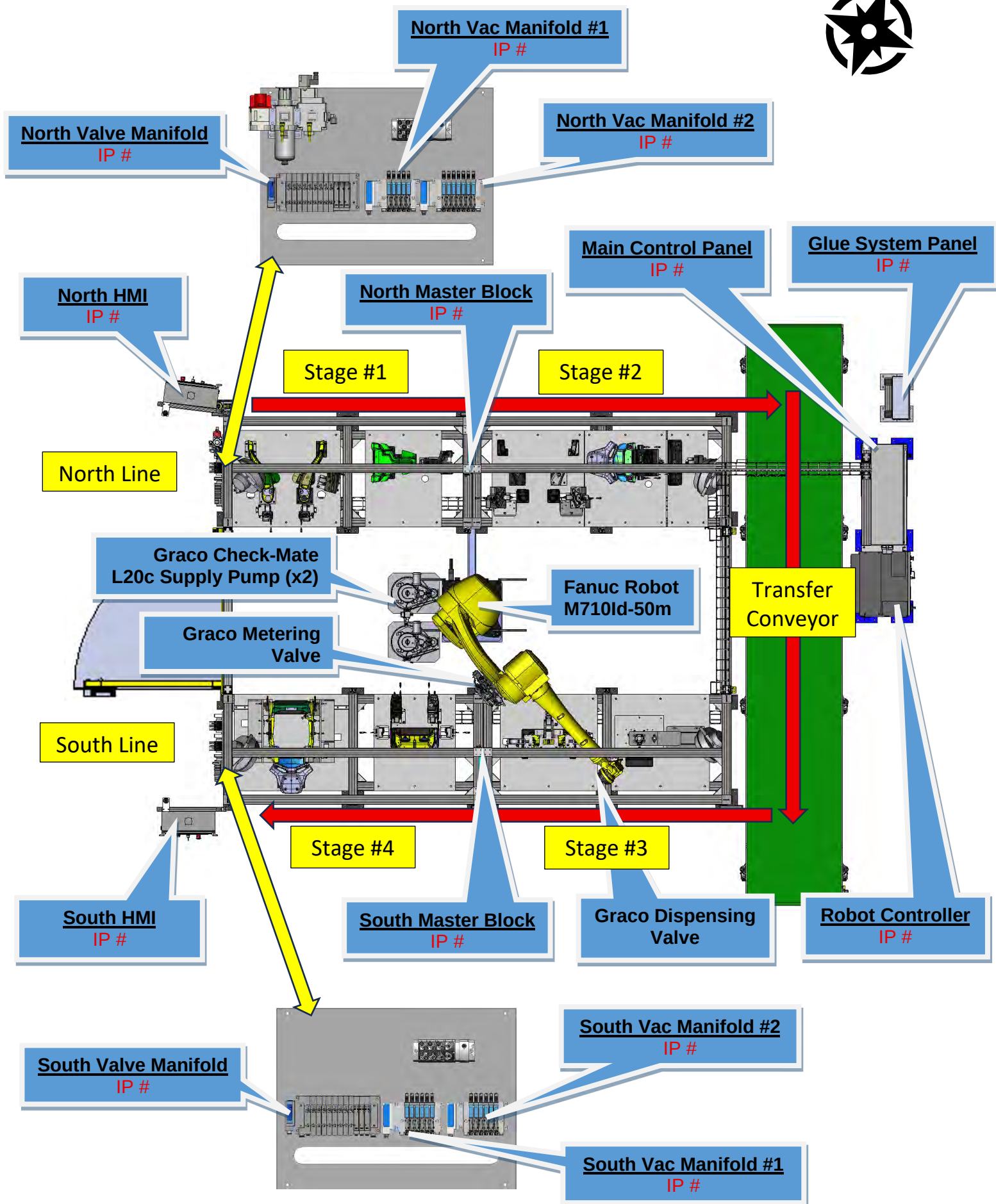
- 8673-800 Cushion Pan:
 - o (x1) cushion pan part present #20
 - o (x1) cushion Left bolster part present #25
 - o (x1) cushion Left bolster vacuum cup
 - o (x1) cushion Left bolster clamp cylinder
 - o (x1) cushion Right bolster part present #26
 - o (x1) cushion Right bolster vacuum cup
 - o (x1) cushion Right bolster clamp cylinder
 - o (x1) cushion Left tab part present (driver side only) #27
 - o (x2) cushion Left tab vacuum cups (driver side only)
 - o (x1) cushion Left tab clamp cylinder (driver side only)
 - o (x1) cushion Right tab part present (driver side only) #28
 - o (x2) cushion Right tab vacuum cups (driver side only)
 - o (x1) cushion Right tab clamp cylinder (driver side only)



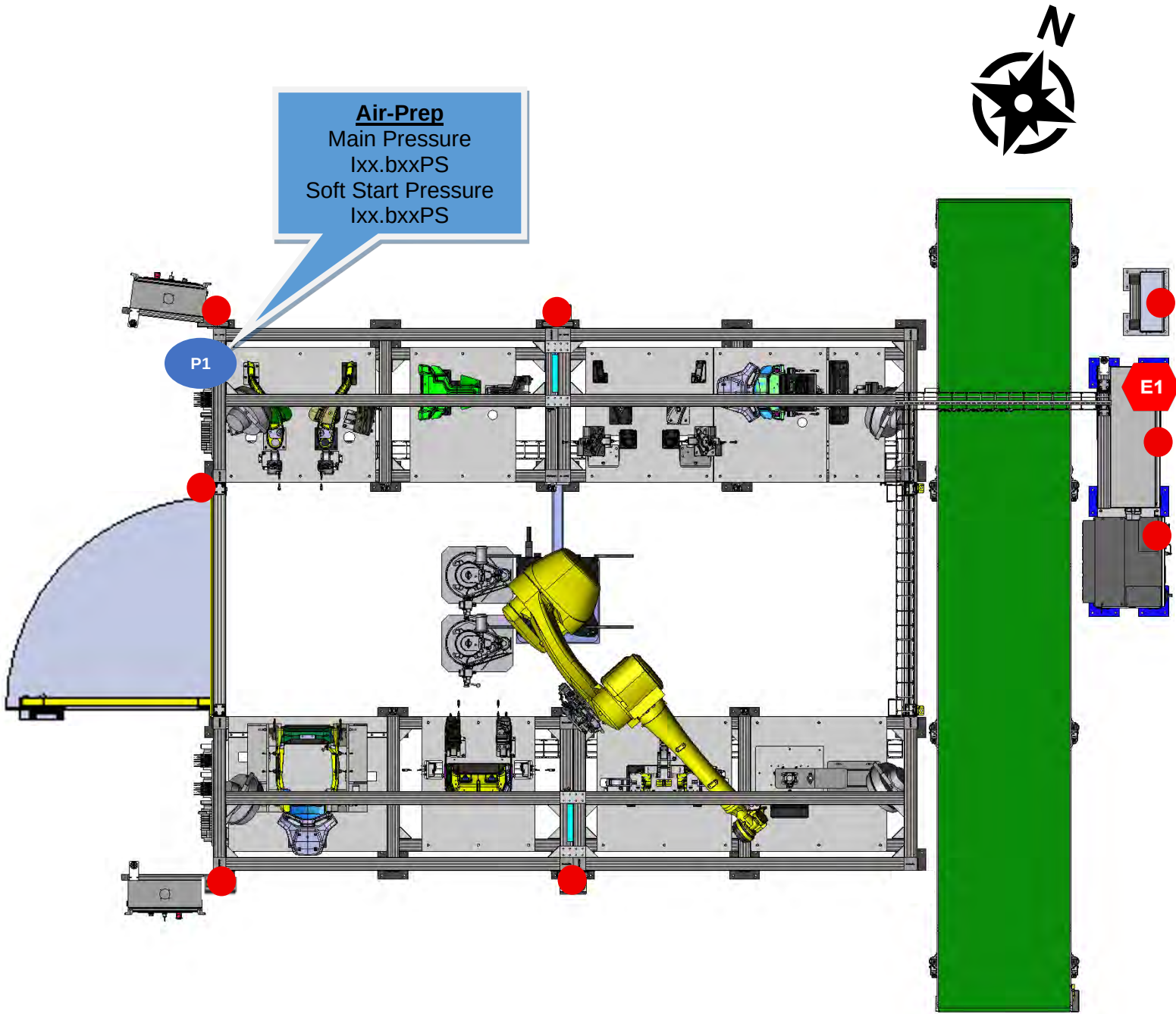
- 8673-900 FSB Final Assembly:
 - o (x1) upper crossmember part present #01
 - o (x1) upper crossmember clamp cylinder
 - o (x1) lower crossmember part present #07
 - o (x1) lower crossmember clamp cylinder
 - o (x1) LH sidemember part present #09
 - o (x4) LH sidemember vacuum cups
 - o (x1) LH sidemember servo clamp
 - o (x1) RH sidemember part present #12
 - o (x4) RH sidemember vacuum cups
 - o (x1) RH sidemember servo clamp



Cell Layout:

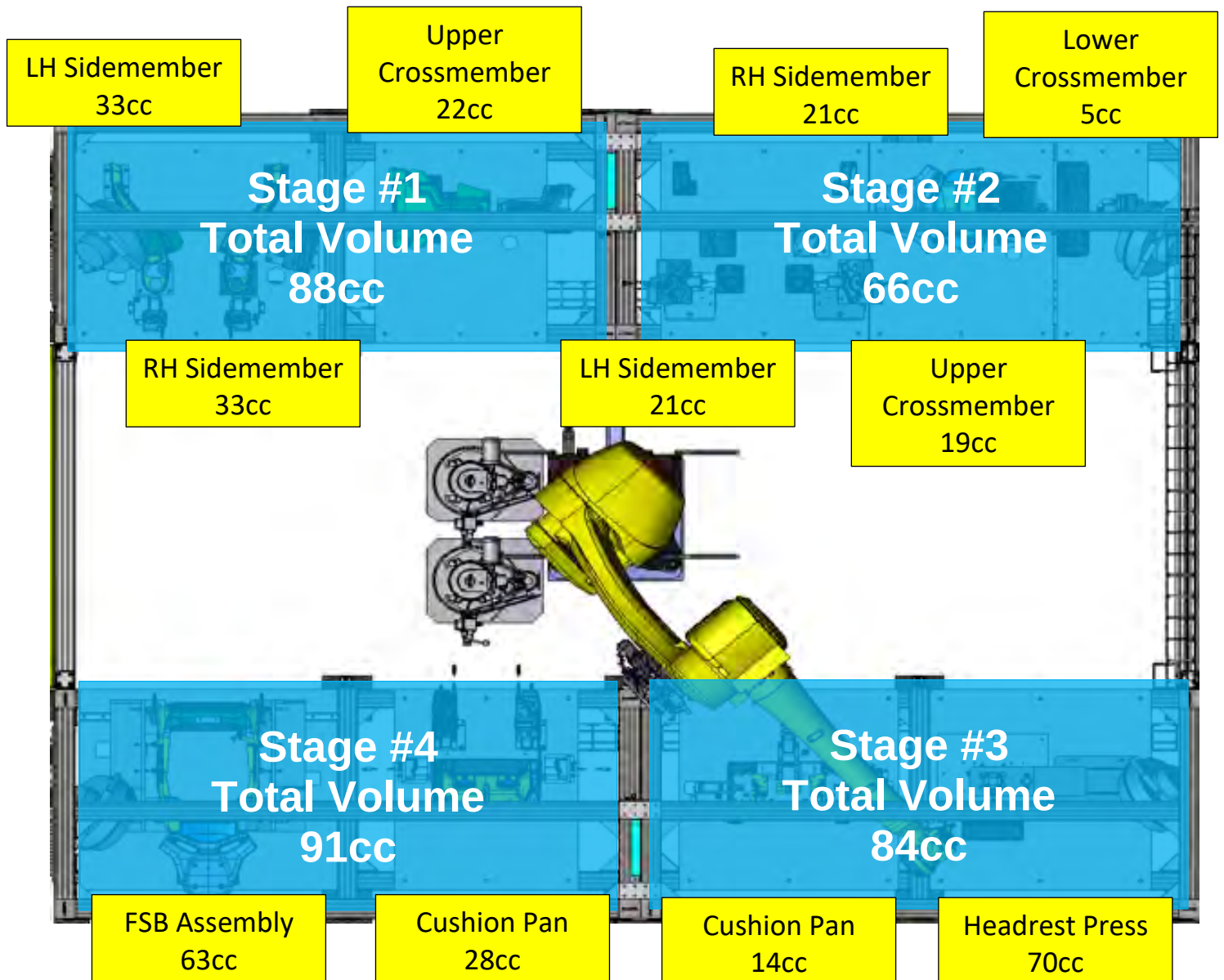
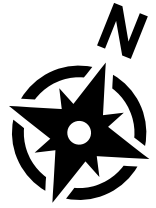


Safety Lockout & E-Stop Layout:



-  Pneumatic Lockout
-  Electrical Lockout
-  E-Stop Locations

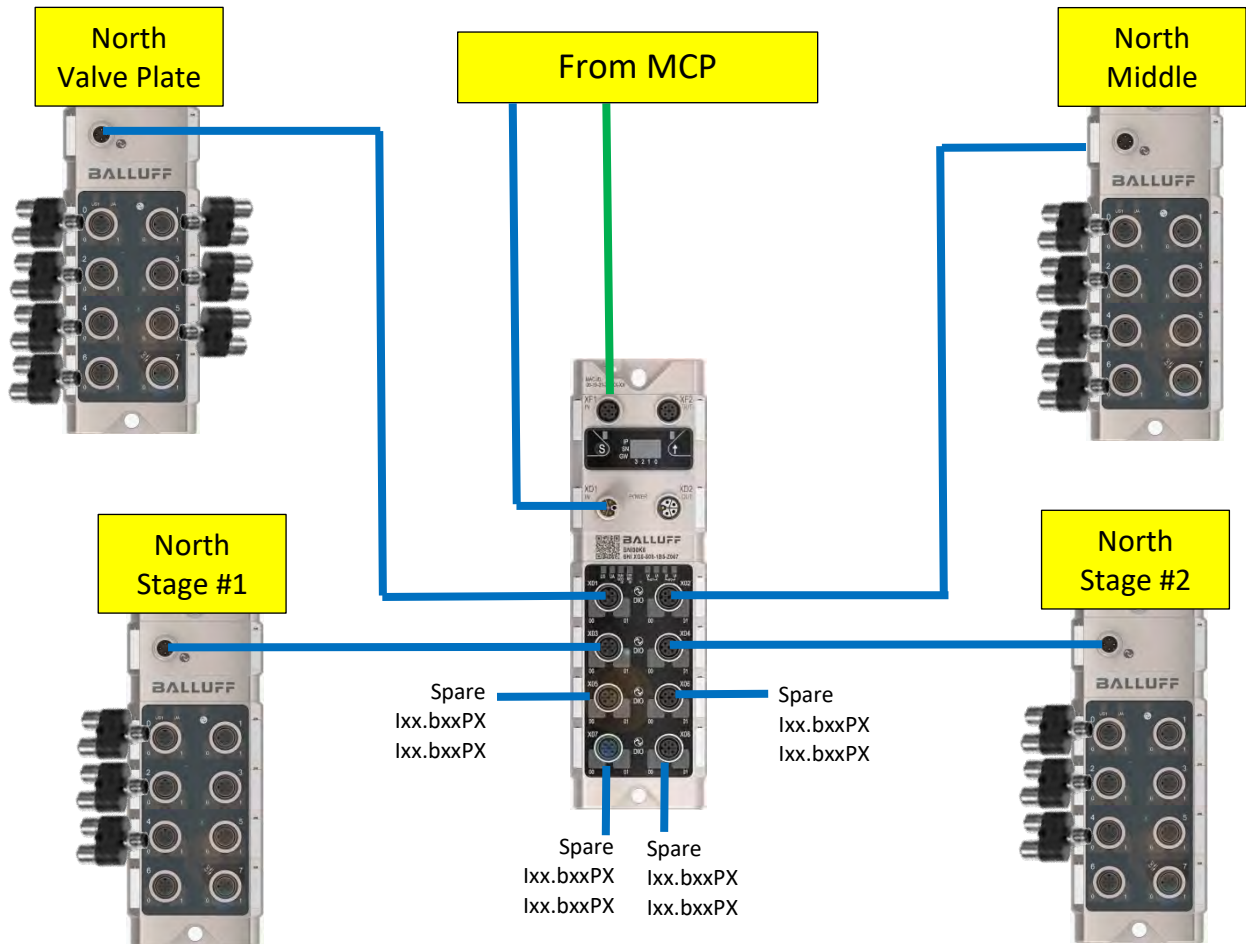
Station Glue Volume:



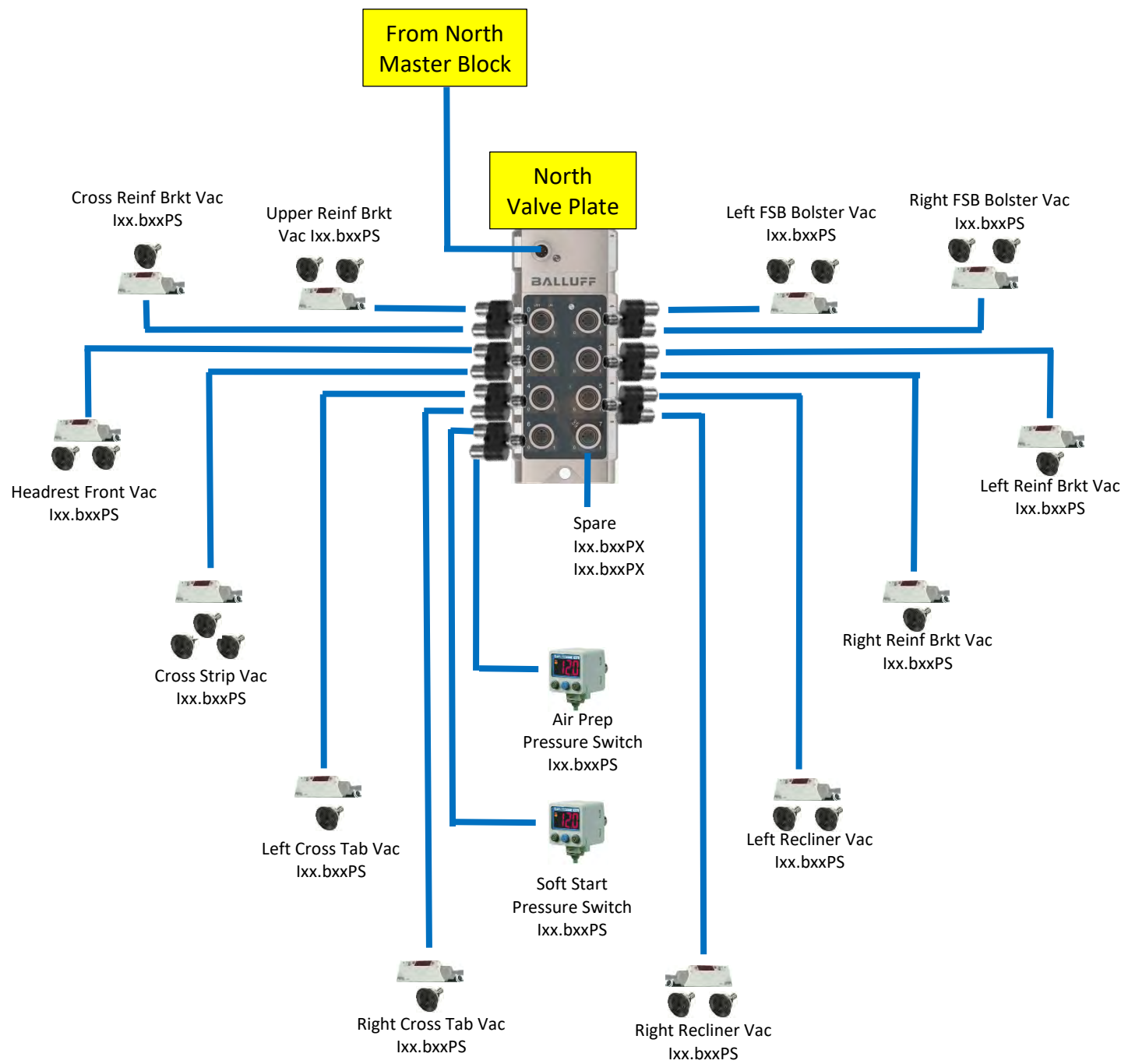
North Line I/O Block Online:

24VDC

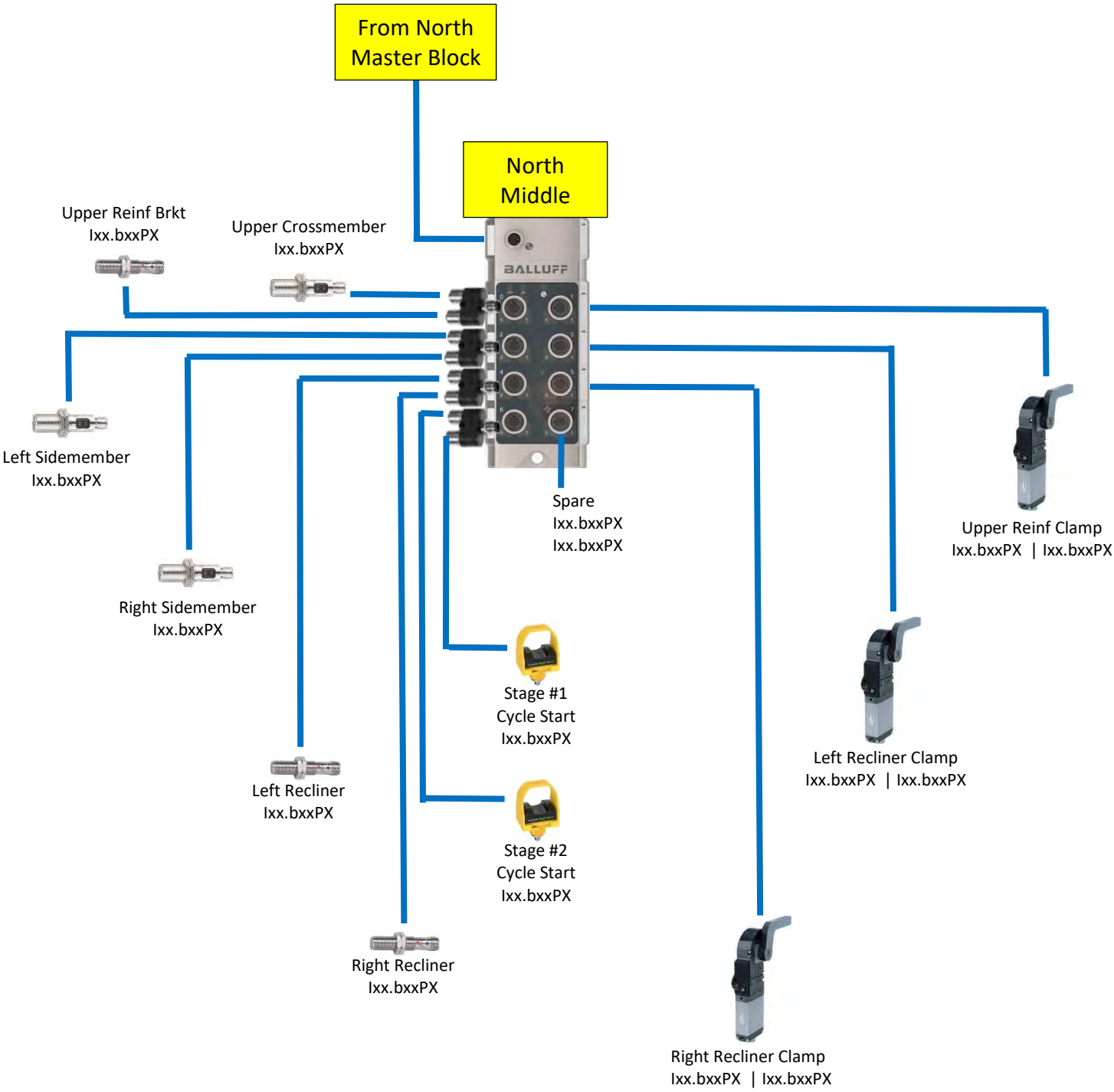
Ethernet/IP



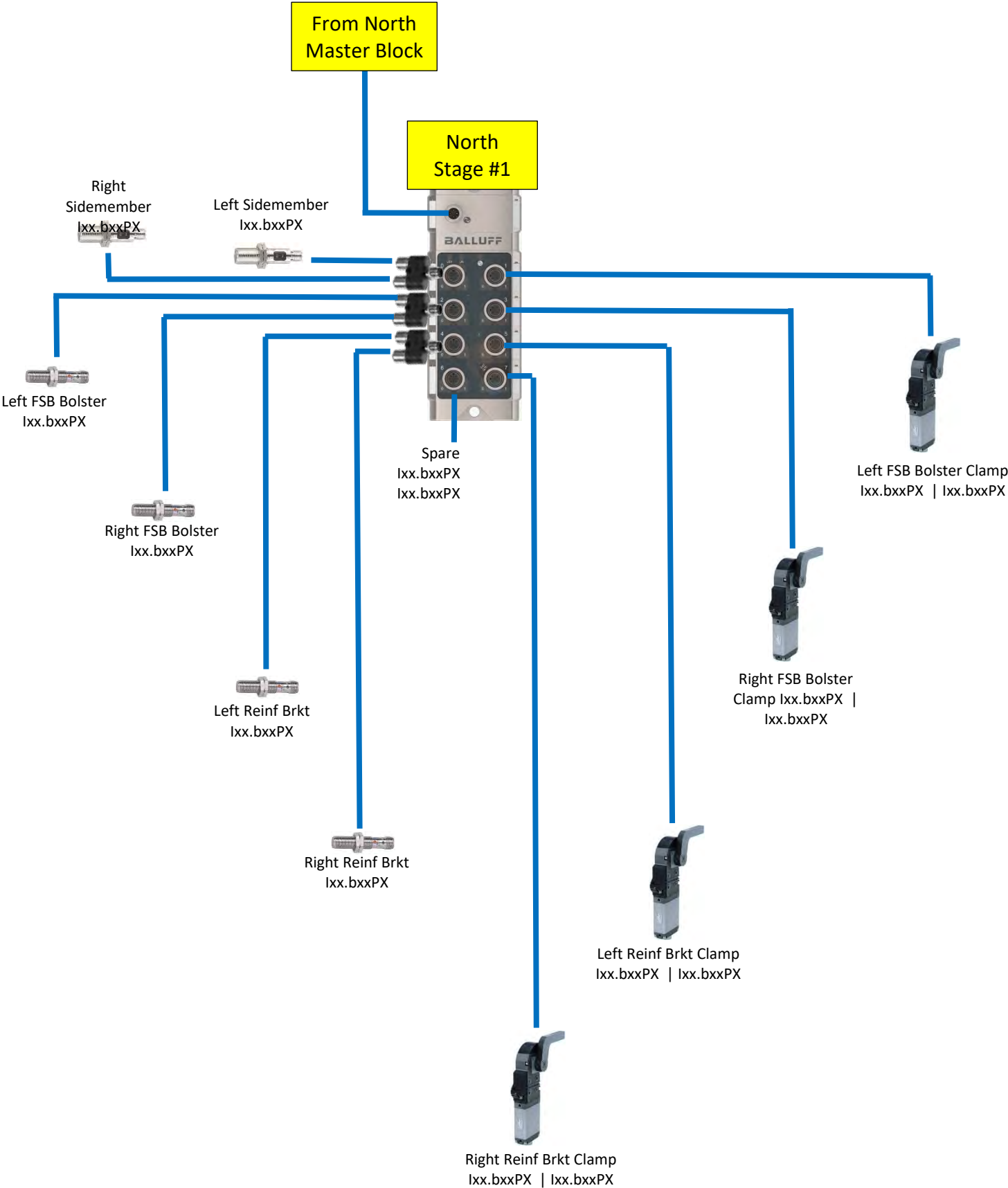
North Line Valve Plate I/O Block Online:



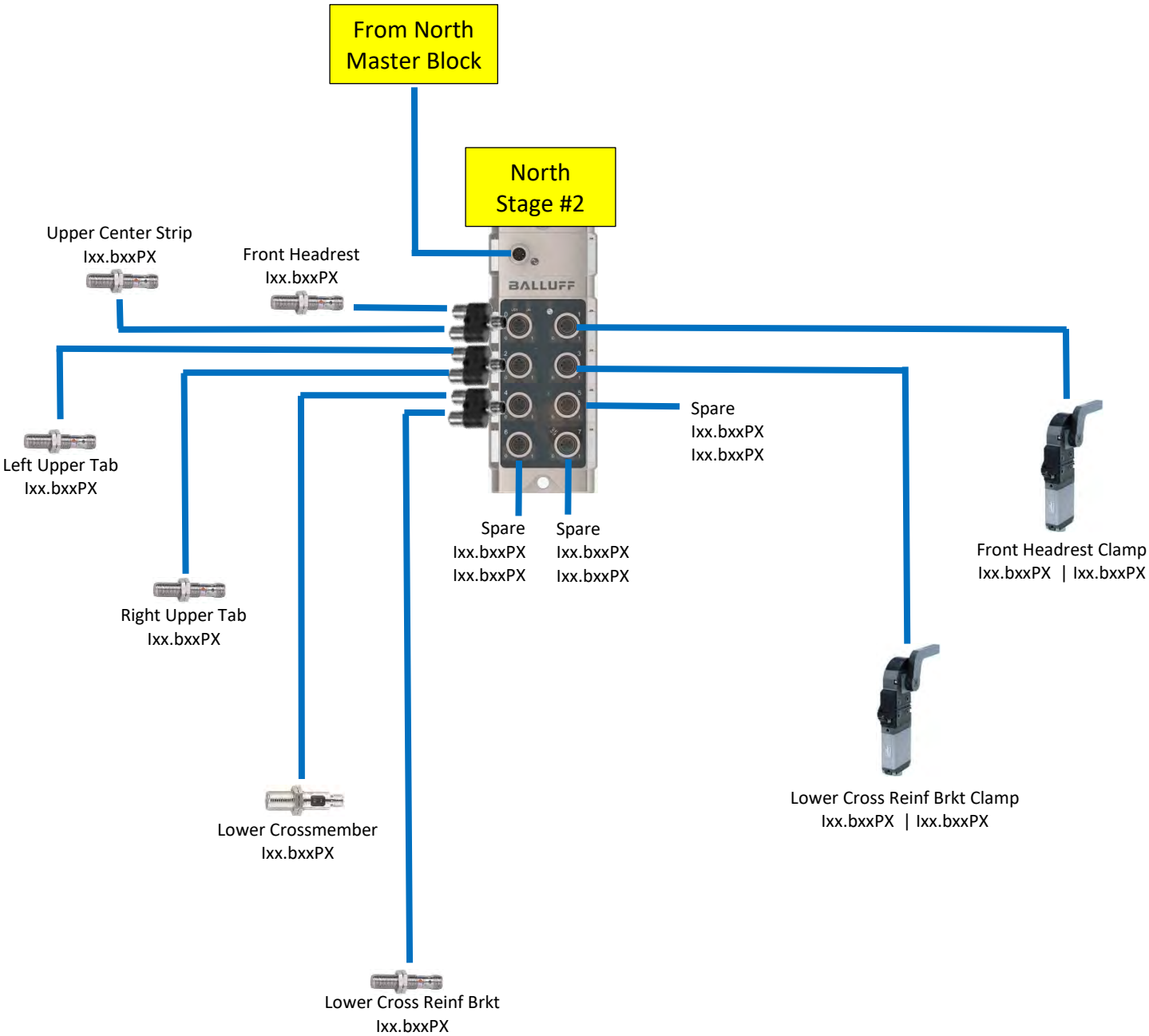
North Line Middle I/O Block Online:



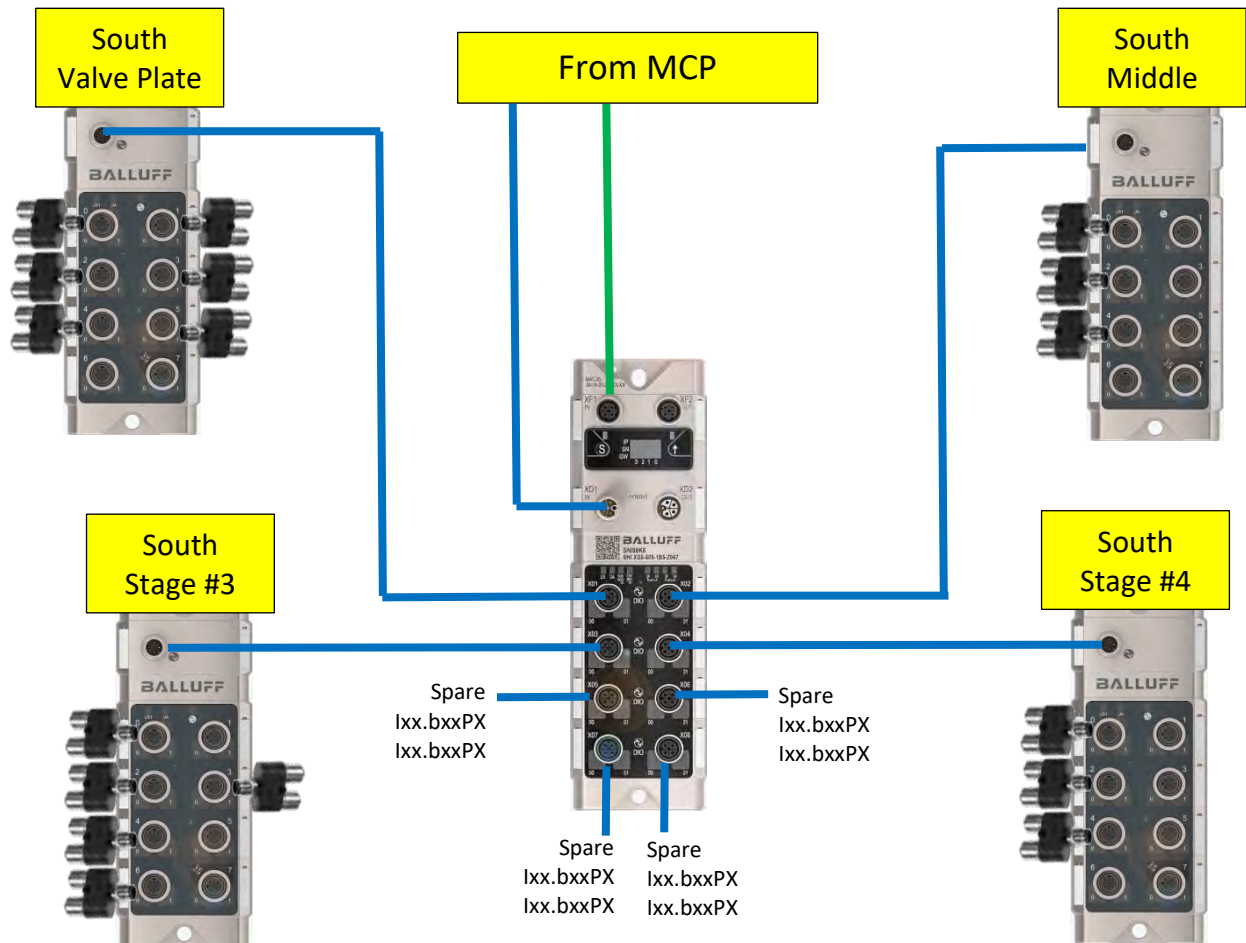
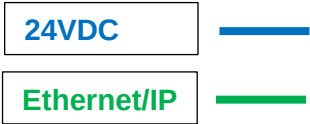
North Line Stage #1 I/O Block Online:



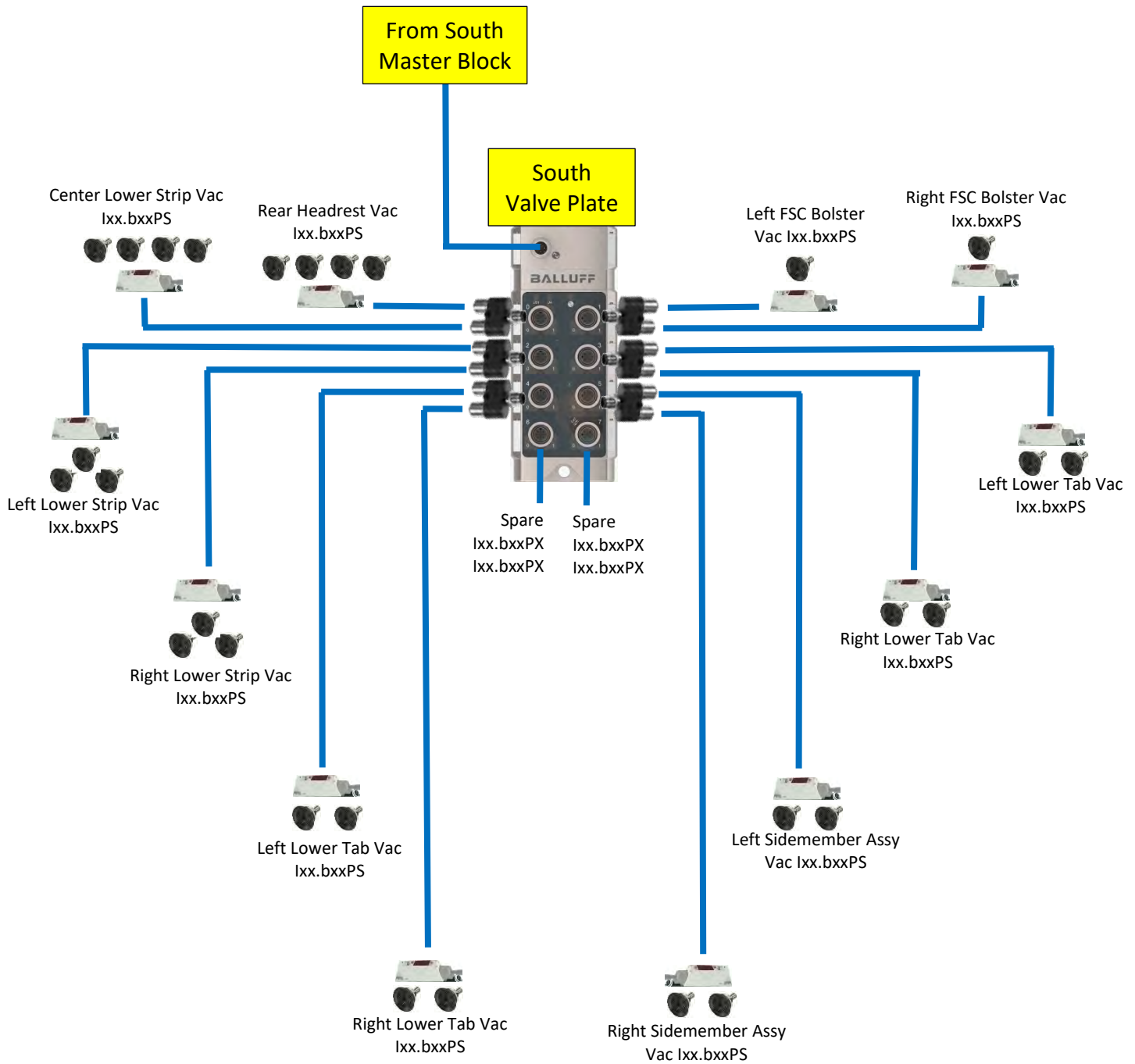
North Line Stage #2 I/O Block Online:



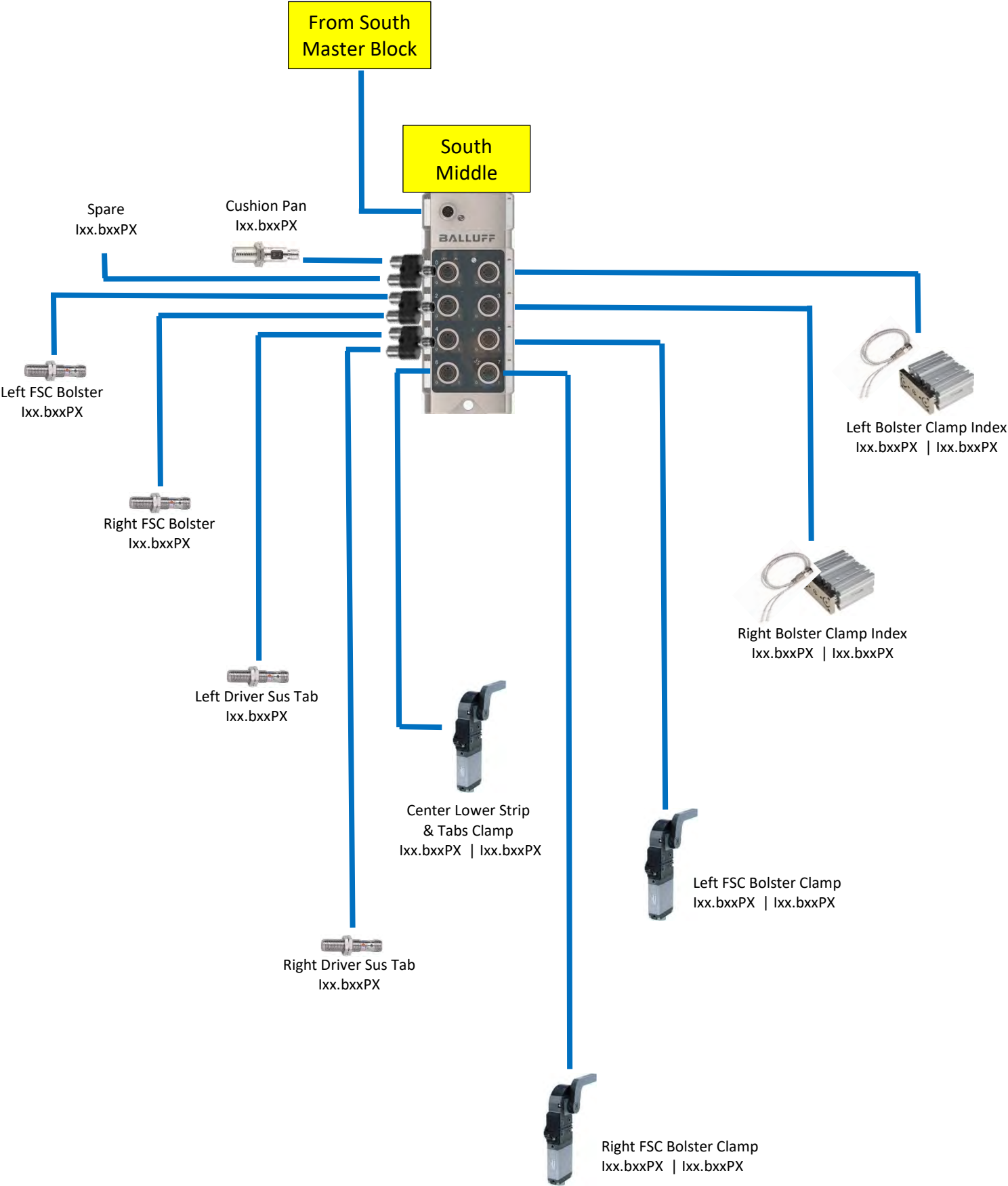
South Line I/O Block Online:



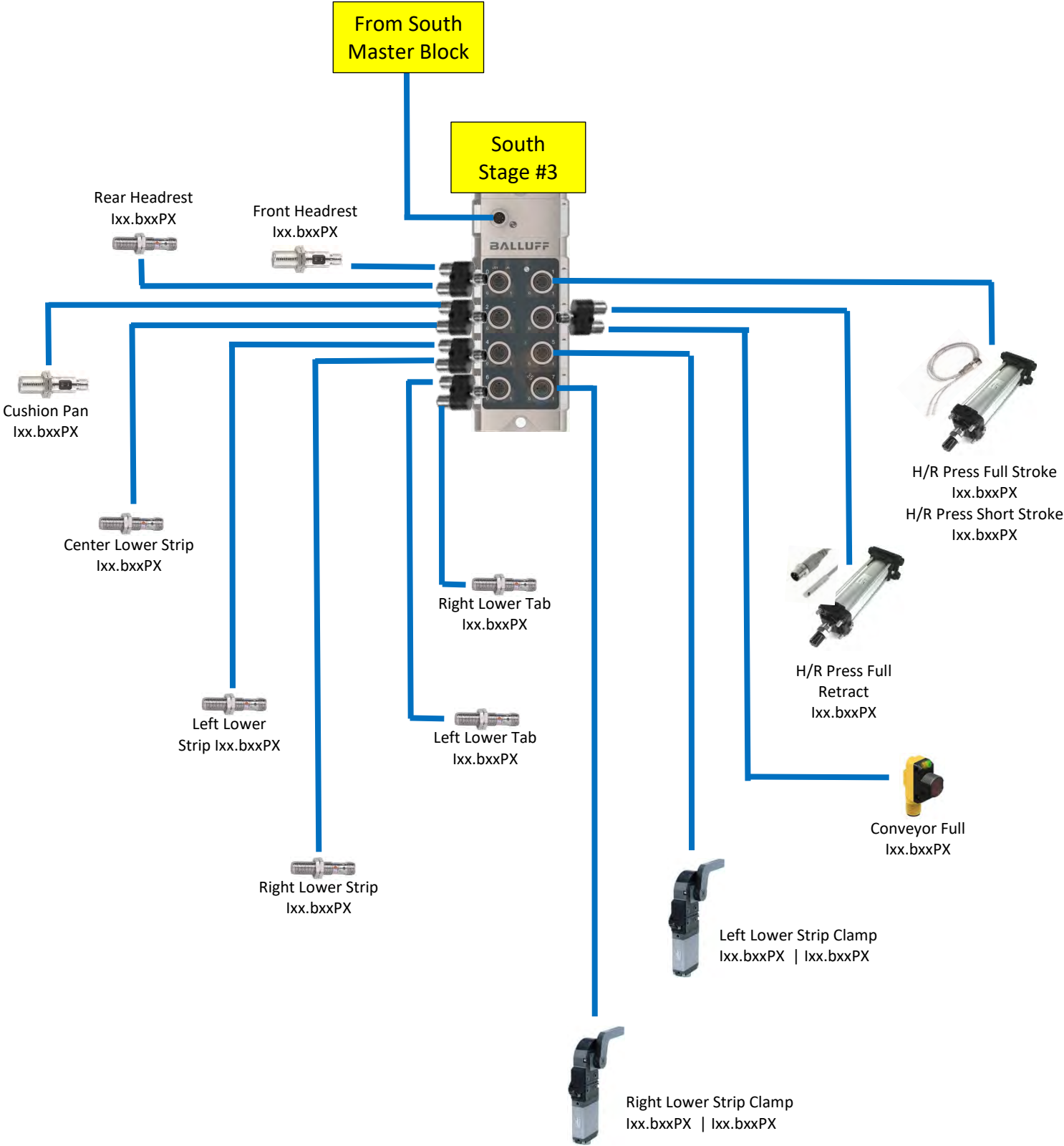
South Line Valve Plate I/O Block Oneline:



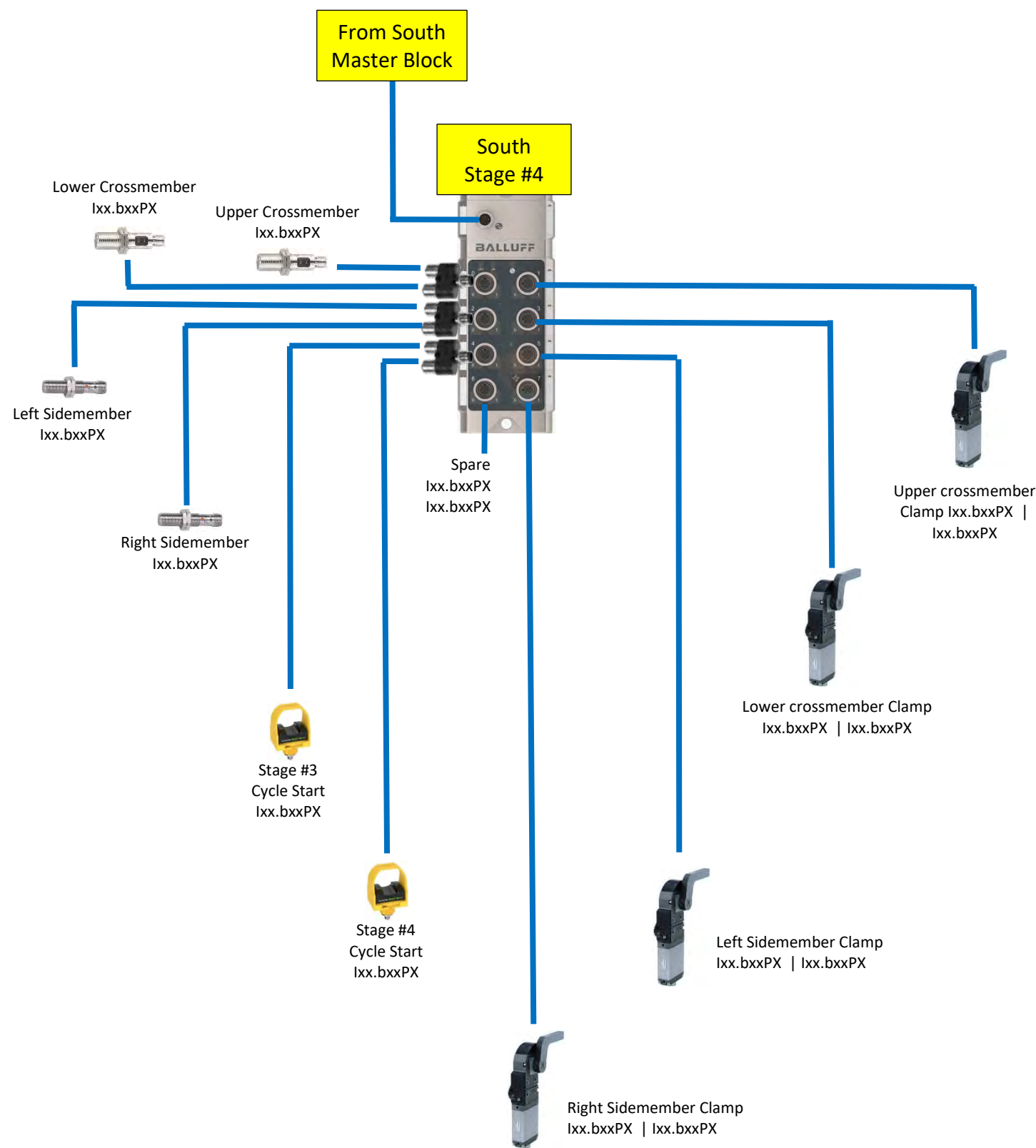
South Line Middle I/O Block Online:



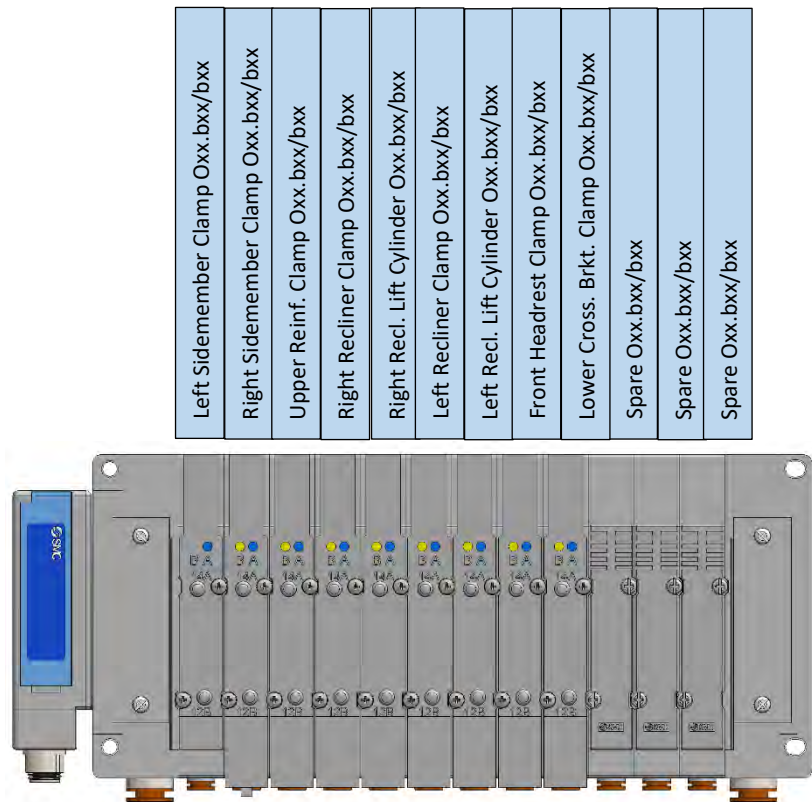
South Line Stage #3 I/O Block Online:



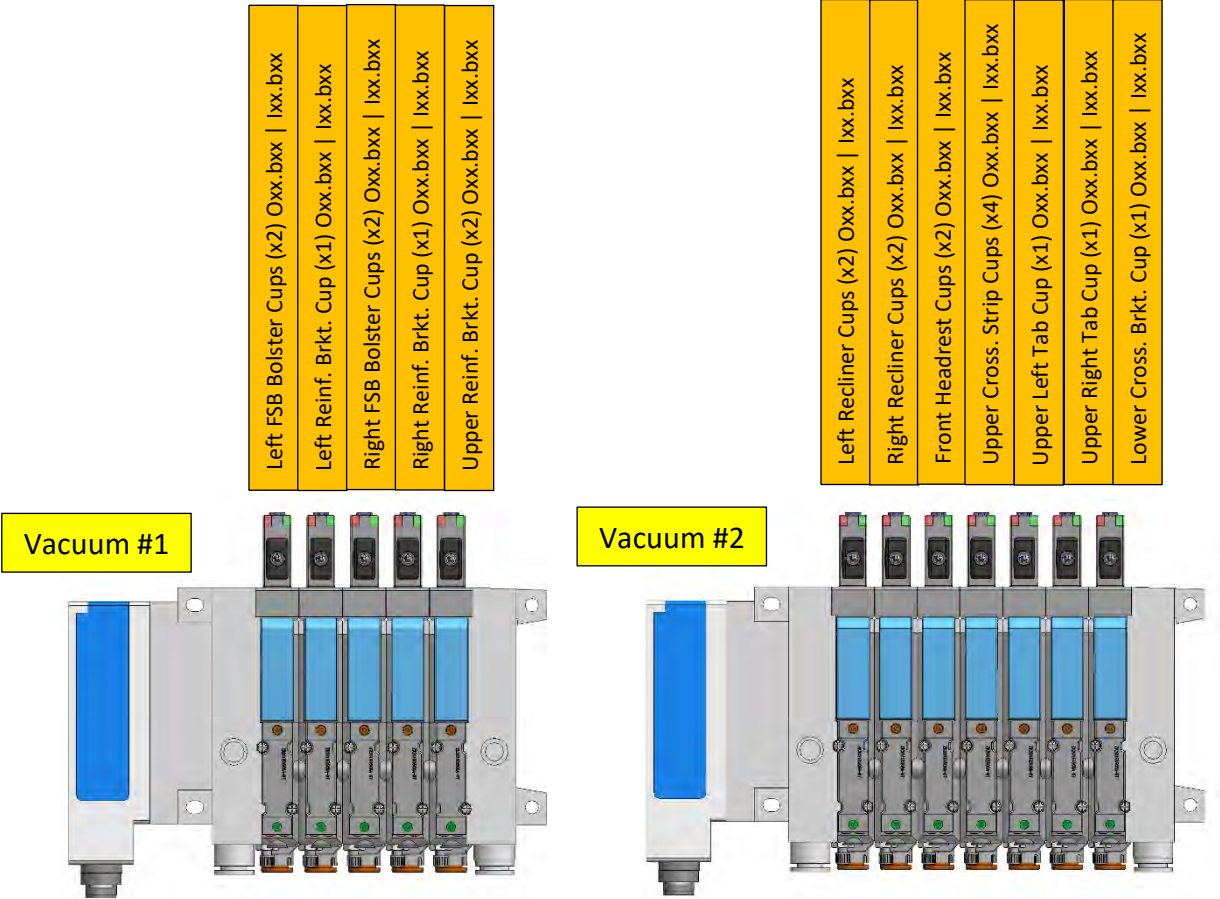
South Line Stage #4 I/O Block Online:



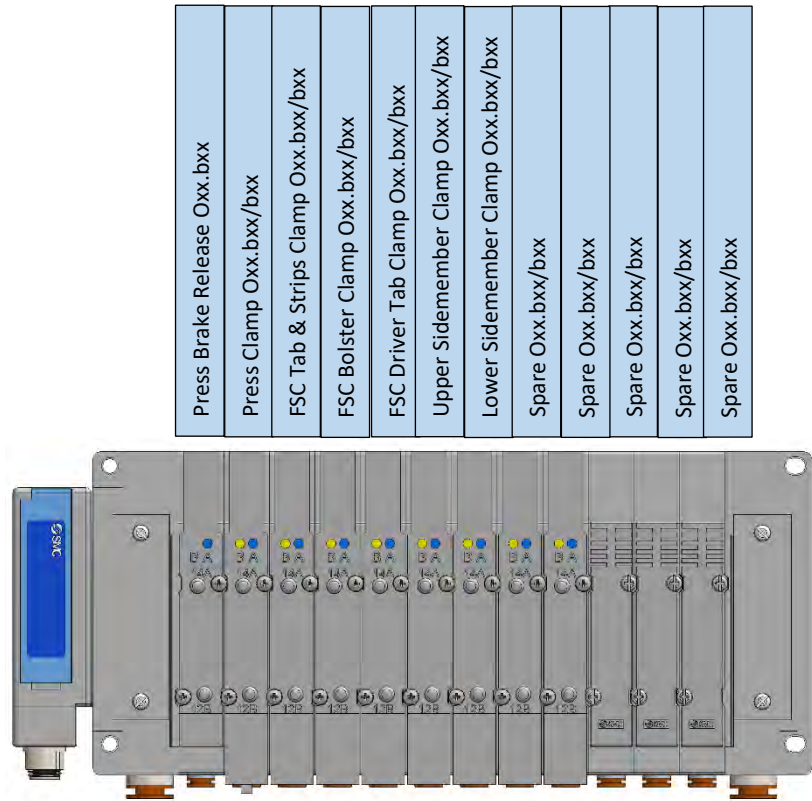
North Valve Manifold Layout:



North Vacuum Manifold Layout:



South Valve Manifold Layout:



South Vacuum Manifold Layout:

